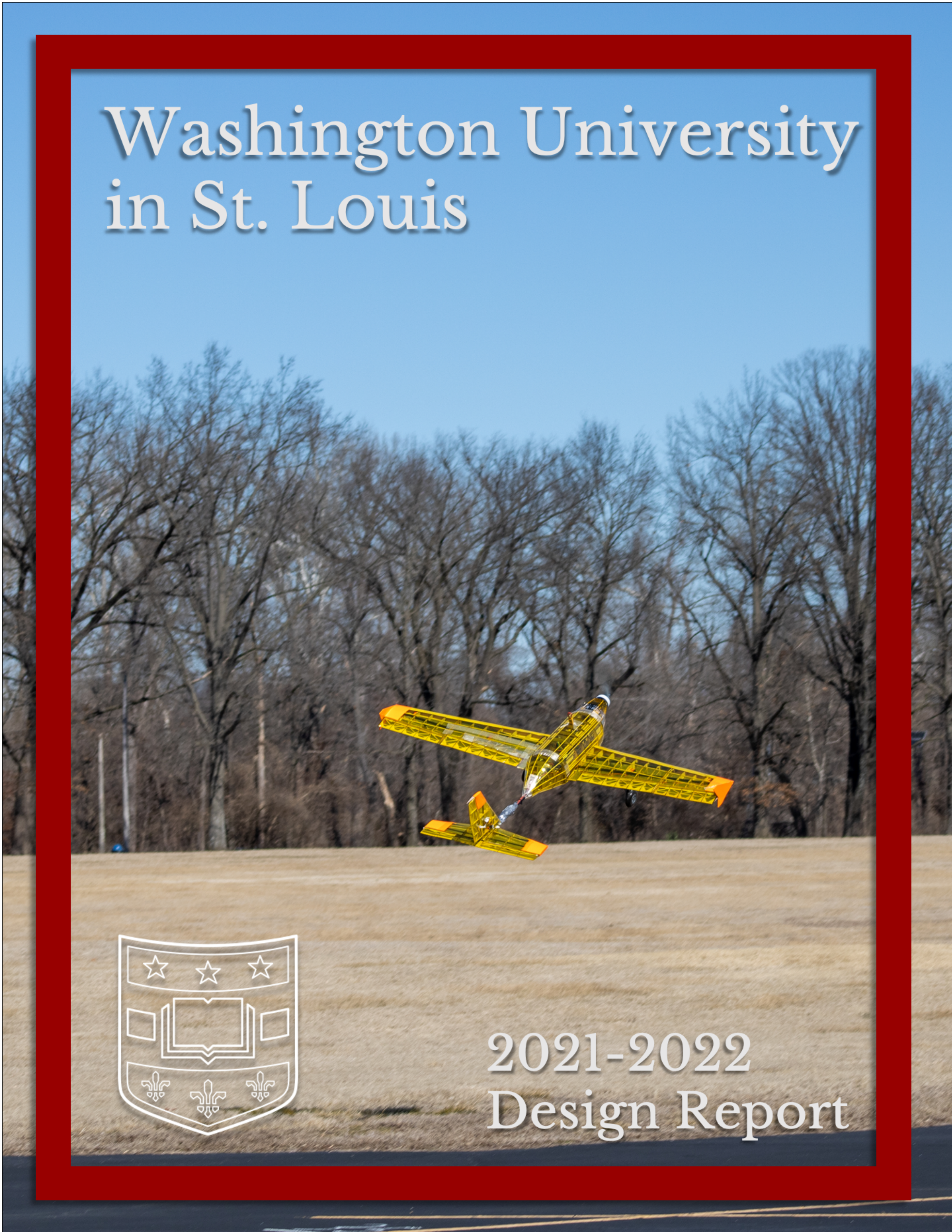
Washington University in St. Louis



2021-2022 Design Report

Contents

Nomenclature

Abbreviations

AIAA	American Institute of Aeronautics and Astronautics
AoA	Angle of attack
AR	Aspect ratio
AUW	All up weight
CA	Cyanoacrylate
CAD	Computer-aided design
CBDM	Component drag build-up method
CF	Carbon fiber
CFD	Computational Fluid Dynamics
CG	Center of gravity
DBF	Design/Build/Fly
FEA	Finite Element Analysis
GM	Ground Mission
HT	Horizontal Tail
HQRS	Cooper-Harper handling qualities rating scale
KM Sim	Kinematic mission simulator
LiPo	Lithium Polymer
M1	Mission 1
M2	Mission 2
M3	Mission 3
NACA	National Advisory Committee for Aeronautics
NiMH	Nickel-metal hydride
RC	Radio-controlled
TOFL	Takeoff field length
T-O	Takeoff
VLM	Vortex lattice method
VT	Vertical Tail
WU	Washington University
WUDBF	Washington University in St. Louis Design/Build/Fly

Symbols

b	Span [m]
c	Chord [m]
C_D	Drag coefficient
C_L	Lift coefficient
Re	Reynolds number
S	Wing Area [m^2]
T	Thrust [kgf]
T/W	Thrust-to-weight ratio
V_{HT}	Horizontal tail volume
v_{max}	Max cruise velocity [m/s]
V_{VT}	Vertical tail volume
W	Weight [kgf]
W/S	Wing loading [kgf/m^2]

1. Executive Summary

The Washington University in St. Louis Design/Build/Fly (WUDBF) 2021-22 Design Report describes the design, testing, and manufacturing of the *Pflyzer*, the team's entry in the American Institute of Aeronautics and Astronautics (AIAA) 2022 Design-Build-Fly competition. For the competition, the aircraft must be able to complete three flight missions and one ground mission. The first flight mission (M1) is a deployment mission that verifies the function of the aircraft. The second mission (M2) is a staging flight; during this mission the aircraft must carry a payload of syringes while flying the course as fast as possible. The third mission (M3) is a vaccine delivery flight involving repeated takeoff/landing with package deployment following each landing. The ground mission (GM) serves as a timed demonstration of the aircraft.

The *Pflyzer* has been designed to meet all mission requirements while maintaining a high level of reliability and minimizing risk. The aircraft consists of a conventional, single motor, taildragger, low-wing aircraft with a dihedral of 5°. These characteristics were chosen during the conceptual design phase in response to the stated requirements. The taildragger configuration gives the plane an initial 11.02° angle of attack on takeoff, which helps decrease takeoff length. The low-wing configuration allows for payloads to be easily installed, which is ideal for the GM. The dihedral was added to improve the stability characteristics of the low-wing configuration. A sensitivity analysis performed during the conceptual design phase indicated that a M2 payload of 40 syringes and a M3 payload of 4 packages was optimal given size constraints and WUDBF's desire for a low-risk design.

During the preliminary design phase, the aircraft was further refined via detailed aerodynamic and propulsion design. Lift, drag, and stability analyses were performed to size the aircraft and set the outer geometry. The wingspan was set at 2400 mm with a wing loading of around 6 kgf/m^2. Package placement was also determined such that center of gravity shifts are minimized following deployment. A propulsion system was then selected that gives the aircraft a 1.52 thrust-to-weight ratio and a predicted maximum cruise velocity of 22.5 m/s.

During the detail design phase, the internal structure of the aircraft was defined. A two stage-deployment mechanism consisting of a servo-controlled release stage and a ramp-based lowering stage was also designed to safely deploy the packages without tripping the shock sensors.

Based on subsequent performance simulations, the aircraft was able to takeoff in less than 12 ft, well below the 25 ft requirement. The aircraft was predicted to be able to complete M2 in 2.192 minutes, well within the time limit and yielding a score of 1.616. For M3, the aircraft was predicted to fly for a total of 3.784 minutes while completing four scoring laps, yielding a predicted score of 1.667. Endurance estimates aligned well with both the M2 and M3 flight times. Adding on the predicted M1 and GM scores, the *Pfylzer* is estimated to score competitively with a total score of 6.283 points out of a possible 7. Thorough ground and flight testing confirmed that the aircraft could meet all predicted performance goals.

2. Management Summary

2.1 Organization

WUDBF consists of 40 members organized into an Administrative Wing and a Technical Wing, both supported by a Faculty Advisor. The Administrative Wing, led by the Administrative President, is responsible for handling resource allocation, purchasing, public relations, and other administrative tasks. The Technical Wing is responsible for the design, manufacture, and testing of the competition aircraft. Within the Technical Wing, five subteams each handle separate aspects of the aircraft design (see Table 2.1 for details). Each subteam is led by a subteam lead and populated with general members. The Technical President oversees the entire Technical Wing to ensure effective collaboration and communication between the different subteams.

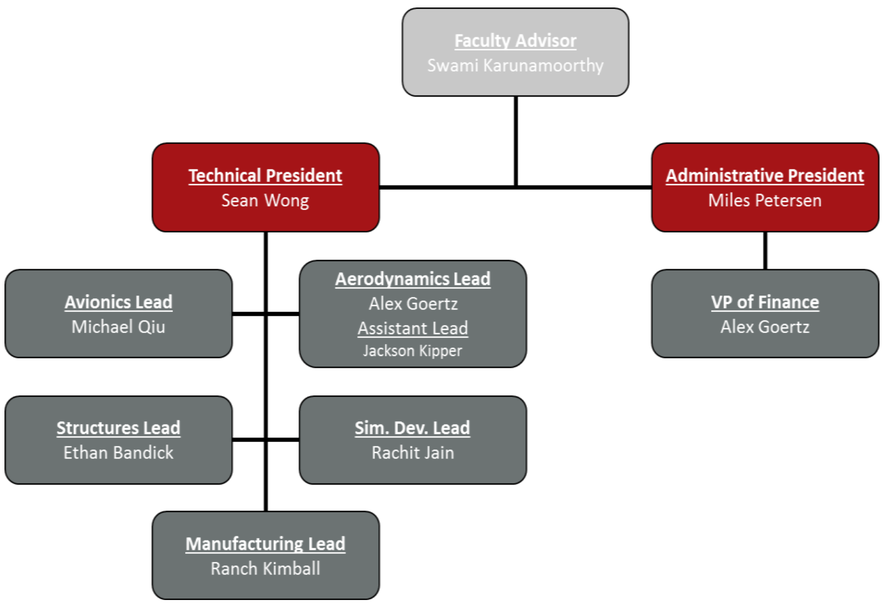


Figure 2.1: Team Hierarchy

Table 2.1: Subteam Roles and Skills

	Aerodynamics	Structures	Manufacturing	Avionics	Simulation Development
Roles	• Design of the exterior geometry of the aircraft • Aircraft sizing	• Design and modeling of the aircraft structure	• Assembly of the aircraft	• Design of the propulsion package • Install of electrical components	• Development of simulation tools • Programming of flight controllers
Skills	• Aircraft stability and control optimization • CFD analysis	• CAD • Structural FEA • Mass properties analysis	• Laser cutting, 3D printing, machining • Balsa RC aircraft construction	• Propulsion theory and analysis • Electrical design • Soldering	• MATLAB/Simulink • Python • Real Flight 8

2.2 Schedule

WUDBF uses a Gantt chart to ensure the team stays on track and meets major deadlines. The Gantt chart for the 2021-22 competition year is shown in Figure 2.2 below. Due to university regulations leading to a delayed start of the Spring semester, WUDBF had to slightly modify the originally planned schedule as January prototype test flights had to be delayed. Despite this, WUDBF still remains on track to arrive at competition with a successful aircraft.

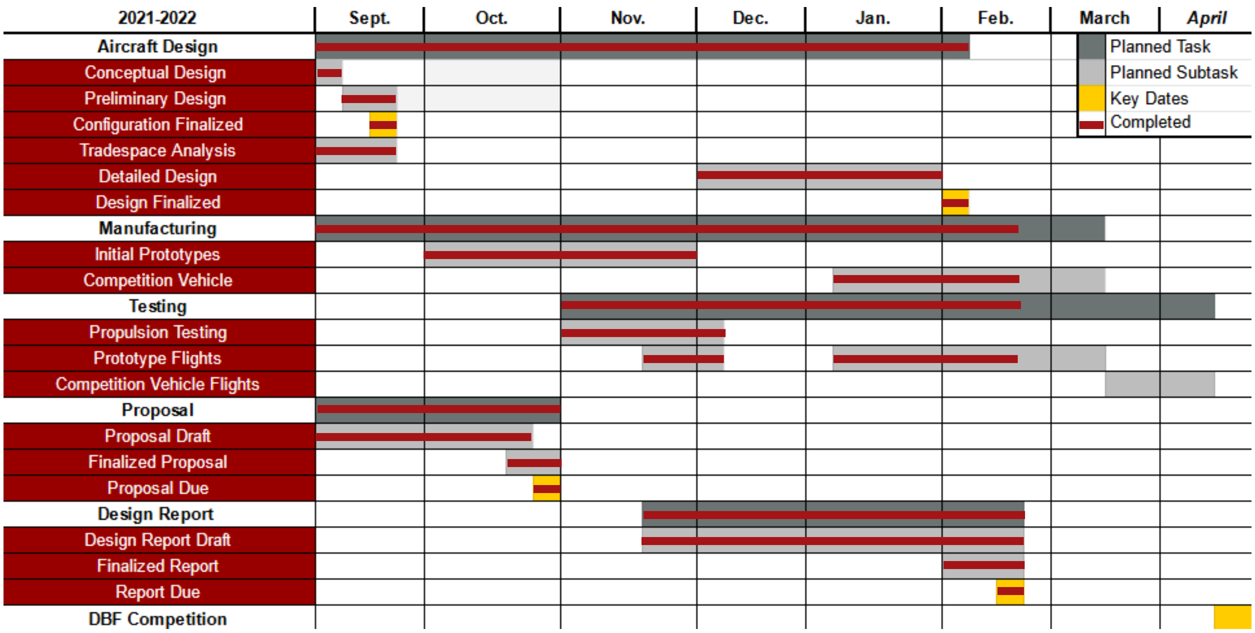


Figure 2.2: WUDBF Gantt Chart for 2021-2022 Competition

3. Conceptual Design

For the 2021-22 DBF competition, the AIAA has set the objective to design and build an aircraft to deliver vaccination components in a humanitarian mission scenario. The aircraft must be capable of transporting vaccination syringes, delivering environmentally sensitive vaccine vial packages, and taking off using short runways (less than 25 feet in length).

Upon receipt of the competition rules, WUDBF began the conceptual design phase starting with identification of key mission requirements and translation of those mission requirements into design requirements, followed by a scoring sensitivity analysis and configuration selection via a series of decision matrices.

3.1 Mission Requirements

A total of four missions are called for in the 2021-22 competition including three flight missions and one ground mission. The competition course layout (used for all flight missions) is shown in the figure below.

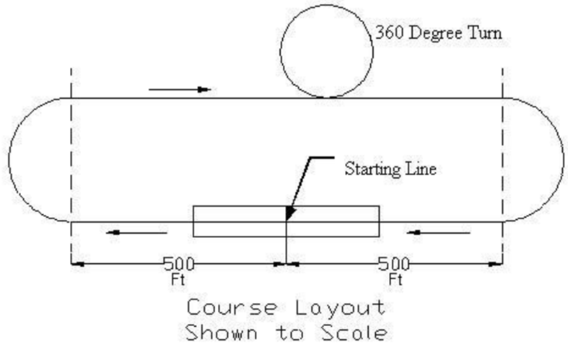


Figure 3.1: Competition Course Layout

3.1.1 Mission 1 – Deployment Flight

Mission 1 takes place with no payload. The aircraft must takeoff within a TOFL of 25 feet and complete 3 laps within a 5 minute flight window. A lap is considered completed when the aircraft passes over the start/finish line. Landing of the aircraft can occur outside of the 5 minute flight window, but a successful landing must be completed in order to receive a score. Scoring for Mission 1 is as follows:

$$M1 = 1.0 \tag{3.1}$$

The maximum achievable score for Mission 1 is **1.0 point**, which is earned upon completion.

3.1.2 Mission 2 – Staging Flight

Mission 2 is the syringe transportation flight. The minimum payload is ten syringes (there is no maximum). Based on syringes obtained by WUDBF, each syringe weighs ~0.65 oz (~0.02 kg). The aircraft with payload must takeoff within a TOFL of 25 feet and complete 3 laps within a 5 minute flight window. Landing of the aircraft can occur outside of the 5 minute flight window, but a successful landing must be completed in order to receive a score. Scoring for Mission 2 is as follows:

$$M2 = 1 + \frac{\left(\frac{\#syringes}{time}\right)_{WUDBF}}{\left(\frac{\#syringes}{time}\right)_{Max}} \tag{3.2}$$

where $\#syringes$ is the number of syringes flown during the mission, $time$ is the time required to complete the mission, and the $WUDBF$ and Max subscripts indicate the $(\#syringes/time)$ score received by WUDBF and the highest $(\#syringes/time)$ received by any team respectively. The maximum achievable score for Mission 2 is **2.0 points** with 1.0 point earned for completion and up to an additional 1.0 point earned for performance.

3.1.3 Mission 3 – Vaccine Delivery Flight

Mission 3 is the vaccine vial package delivery flight. The minimum payload is one package. The maximum payload is the maximum number of syringes flown in a successful Mission 2 divided by ten (rounded down to the nearest whole number). Each package consists of a wood block containing three 25G shock sensors. The weight of each package will be ~8.00 oz (~0.23 kg). The aircraft with payload must takeoff within a TOFL of 25 feet. There is a 10 minute flight window for Mission 3 in which to complete as many laps as possible. Following completion of one flying lap, the aircraft must land on the runway and taxi to the designated vaccine vial package drop area. Once within the designated area, the aircraft must remotely deploy one package. After successfully deploying a package, the aircraft must then taxi across the start/finish line and takeoff again within 25 feet, complete another flying lap, and deploy a single package until all packages are deployed or the flight window expires. Landing of the aircraft can occur outside of the 10 minute flight window, but a successful landing must be completed in order to receive a score. Scoring for Mission 3 is based on the number of successful vaccine vial packages deployed within the designated drop area without tripping the shock sensor:

$$M3 = 2 + \frac{(\#successful\ deployments)_{WUDBF}}{(\#successful\ deployments)_{Max}} \tag{3.3}$$

where $\#successful\ deployments$ refers to the number of packages successfully deployed (i.e. without tripping the shock sensor) and the $WUDBF$ and Max subscripts indicate the $(\#successful\ deployments)$ achieved by WUDBF and the highest $(\#successful\ deployments)$ received by any team respectively. The maximum achievable score for Mission 3 is **3.0 points** with 2.0 points earned for completion and up to an additional 1.0 point earned for performance.

3.1.4 Ground Mission – Operational Demonstration

The Ground Mission serves as a timed ground demonstration of the aircraft's Mission 2 and 3 capabilities. The mission begins with the aircraft in the "mission box" with the maximum declared number of vaccine vial packages and the required number of syringes (ten times the maximum number of packages). The mission is divided into three phases. In the first phase, the mission official will say "GO" upon which time will start. An assembly crew member will then run from behind the start/finish line to the mission box. Once there, they will load the full Mission 2 payload (the syringes) and run back to the start/finish line; time stops when they cross the line. The second phase starts when the official says "GO" again and time restarts. The assembly crew member will then run back to the mission box, remove the Mission 2 payload, install the full Mission 3 payload (the packages), and run back to the line where time will stop. The timed portion concludes with the second phase. In the third phase, all packages will then be deployed remotely one at a time to validate functional performance. All packages must be deployed for the Ground Mission to count. Scoring is as follows:

$$GM = \frac{(time)_{Min}}{(time)_{WUDBF}} \tag{3.4}$$

where $time$ is the time required to complete the timed portion of the Ground Mission (phases 1 and 2 as described above) and the $WUDBF$ and Min subscripts indicate the time achieved by WUDBF and the fastest ground mission time achieved by any team respectively. The maximum achievable score for the Ground Mission is **1.0 points** based on performance.

3.1.5 Total Competition Score

The Mission 1, Mission 2, Mission 3 and Ground Mission scores will be combined to give the Total Mission Score:

$$\textit{Total Mission Score} = M1 + M2 + M3 + GM \tag{3.5}$$

The maximum achievable Total Mission Score is **7.0 points** with 4.0 points earned based on completion and 3.0 points earned based on performance. The Total Mission Score is then multiplied by the Design Report Score to give the Total Competition Score (SCORE):

$$\mathrm{SCORE} = \textit{Design Report Score} \times \textit{Total Mission Score} \tag{3.6}$$

3.1.6 General Mission Requirements

In addition to the requirements specific to each mission, there are also a number of other rules of note that apply to the competition aircraft during all missions:

- The aircraft cannot exceed 8 feet in any linear dimension.
- All payloads must be carried internally to the aircraft. No part of any payload can be part of or protrude outside of the airplane external surfaces or features.
- Any device used to load, store, or deploy a payload must remain installed on the aircraft during all flight missions and be included during all phases of the Ground Mission.
- The aircraft may utilize either NiCad/NiMH OR Lithium Polymer (LiPo) batteries for propulsion, but the total propulsion power total stored energy cannot exceed 100 watt-hours.
- The flight missions must be flown in order; a new mission cannot be flown until the team has obtained a successful score for the preceding mission.

3.2 Translation to Design Requirements

Based on the mission requirements outlined above, WUDBF formulated design requirements for each mission. These design requirements are broken down in the table below.

Table 3.1: Translation of Mission Requirements into Design Requirements

Mission	Mission Requirement	Design Requirement
All	25 ft TOFL	Low W/S, high $C_{L,max}$, high T/W
M2	Minimize lap time	High cruise velocity, high T/W, low drag
	Maximize payload	Large wing area, sufficient internal volume
M3	Maximize payload	Large wing area, sufficient internal volume
	Remote payload deployment	Electronic deployment mechanism
GM	Minimize loading time	Optimized payload storage configuration, easy access to internal storage
General	Complete every mission	Low risk design, high stability
	Max propulsion energy of 100 watt-hours	High efficiency, constrains maximum propulsion power
	Max linear dimension of 8 ft	Constrains wingspan

3.2.1 Analysis of Design Requirements

TOFL: The short takeoff imposes a number of major design requirements. In order to achieve the 25 ft TOFL, the aircraft must have some combination of low wing-loading, high $C_{L,max}$, and high thrust-to-weight ratio. These are critical design requirements since no flight mission can be completed without meeting the 25 ft TOFL requirement.

Mission 2: In order to maximize Mission 2 score, the aircraft must be designed to maximize payload while minimizing lap time. These two mission requirements lead to somewhat opposing design requirements. Increasing the payload requires an increase in the overall size of the aircraft; the wing area must increase to maintain a reasonable wing-loading. This however increases the drag on the aircraft, decreasing the cruise velocity. Thus, the aircraft must achieve a balance between these two sets of design requirements in order to maximize score.

Mission 3: Scoring in Mission 3 is based entirely on the number of successful deployments. Thus, the primary design requirement is for a large wing area and internal volume to maximize the number of packages that can be carried. It is important to note, however, that designing for Mission 3 impacts Mission 2 as well, since designing for increased Mission 3 payload would also decrease Mission 2 cruise velocity. Increasing payload also impacts takeoff ability.

Ground Mission: In order to minimize the Ground Mission time and increase score, the payload storage configuration must be optimized to allow for easy loading/unloading of Mission 2 and 3 payloads. For the same reason, the fuselage of the aircraft should allow easy access to the storage compartment via a hatch or door.

Minimal Risk: From previous experience and the results from past competitions, WUDBF determined that only a fraction of teams successfully complete every flight mission. Successfully completing every mission is critically important in order to maximize score, especially for this year's competition where over half of the points are awarded for completion. Consequently, WUDBF prioritized designing a reliable, stable aircraft that minimizes risk.

Constraints: The 100 watt-hour maximum propulsion energy limit imposes a hard constraint on the design of the aircraft. This limits the maximum power of the propulsion system and creates a need for a high efficiency design. The 8 ft maximum linear dimension constrains the wingspan of the aircraft.

3.3 Scoring Sensitivity Analysis

Based on the design requirements formulated above, WUDBF determined that there was no single obvious design direction. Prioritizing Mission 2 cruise velocity to minimize lap time drives the design towards a smaller, faster aircraft. Prioritizing increased payload in Missions 2 and 3 drives the design towards a larger, slower aircraft. Therefore, WUDBF conducted a scoring sensitivity analysis to determine which design direction maximizes total score.

3.3.1 Scoring Model

To construct a scoring model, WUDBF first identified the key scoring variables in the scoring equations: M2 time, number of syringes flown in M2, number of deployments in M3, and GM time. A sensitivity analysis could be conducted using all four scoring inputs. However, WUDBF determined that these variables are not all independent and could be linked together. Increasing M3 payload requires increasing the M2 payload (since the number of syringes flown in M2 must be at least ten times the number of packages flown in M3). Increasing the payload increases the weight of the aircraft, which in turn impacts the maximum cruise velocity in M2 and M2 time. GM time is also affected, since increasing the payload increases the time required to load/unload it.

Thus, WUDBF constructed a scoring model using the **number of deployments in M3** as the sole input variable. Various equations were then developed to link it to M2 time, the number of syringes flown in M2, GM time, and ultimately Total Mission Score. A diagram depicting the full scoring model is shown in Figure 3.2. In order to implement the model, a number of simplifying assumptions had to be made. These assumptions and their associated justifications are detailed in Table 3.2.

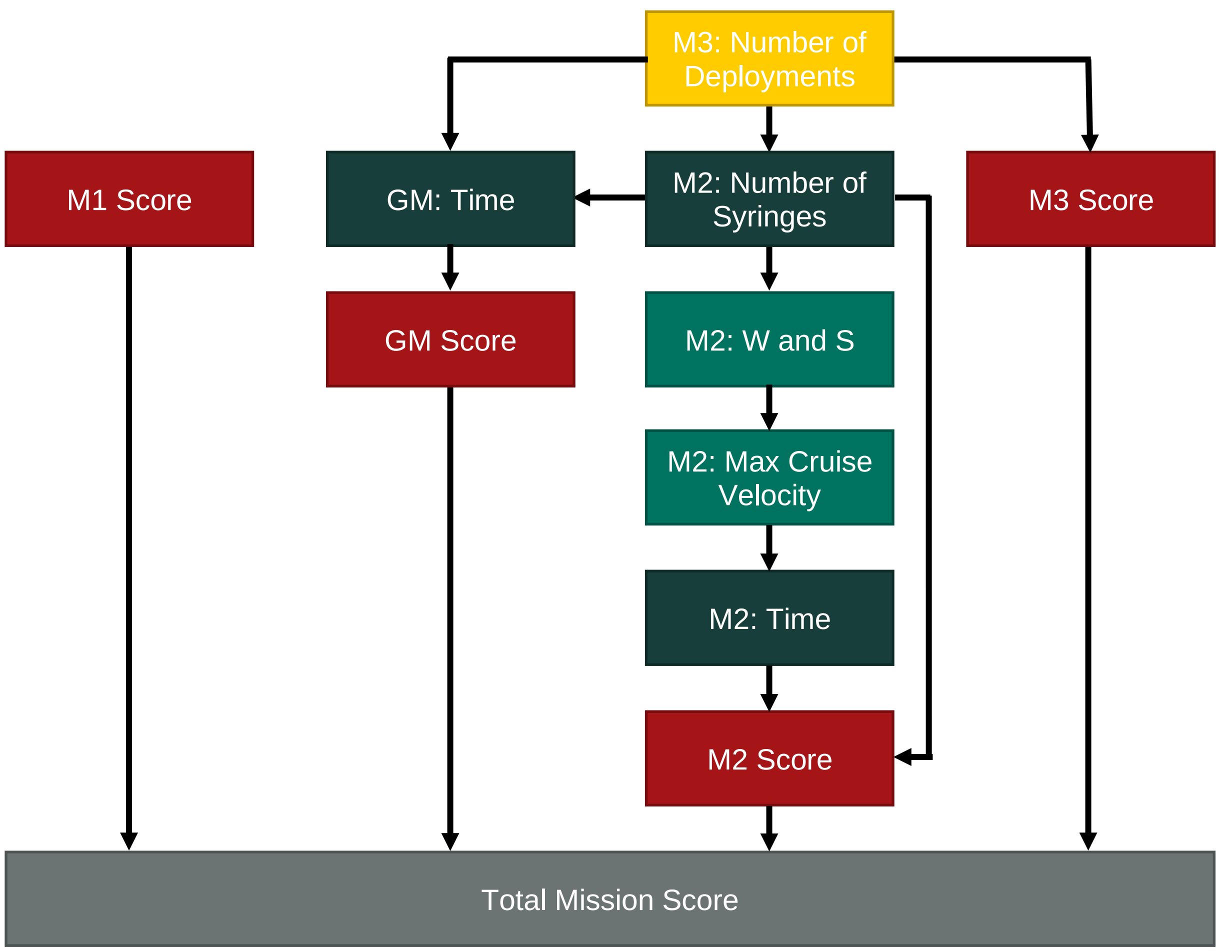


Figure 3.2: Scoring Model Flowchart

Table 3.2: Table of Key Assumptions and Justifications for Scoring Model

Assumption	Justification
10 syringes weigh the same as 1 package and take up approximately the same volume.	Based on empirical testing done by WUDBF.
The number of syringes carried in M2 is 10x the number of deployments in M3.	Based on the minimum amount required by the rules. Theoretically, more syringes could be flown in M2 than required, but for simplicity that possibility is ignored in this model. Furthermore, since 10 syringes weighs about the same and takes up the same volume as 1 package, it does not make sense to design an aircraft that can carry a significantly greater M2 payload than M3 payload.
GM time increases linearly with the number of syringes/deployments.	Based on empirical testing done by WUDBF.
M2 all-up weight (W) increases linearly with the number of syringes carried in M2.	Based on experience from previous DBF competitions. The amount of structural mass required to support each additional payload is approximately constant.
The aircraft has a wing-loading of 6.5 kg/m^2	Based on experience from previous DBF competitions. An aircraft with this wing-loading is likely to be able to take-off within 25 ft.
The aircraft has an AR of 5.5.	Based on minimal risk requirement. It is generally agreed that an AR of between 5:1 and 6:1 is ideal for an easy flying aircraft [1].
Max cruise velocity can be estimated from all-up weight (W) and wing area (S).	Max cruise velocity can be determined by iteratively solving the following equation [2, p. 878]: $\rho S C_{Dmin} {V_{max}}^3 = \eta P + \sqrt{(\eta P)^2 - 4W^2 {V_{max}}^2 C_{Dmin} k}$
Lap length is approximately 2500 feet.	Based on course diagram.

3.3.2 Results of Scoring Sensitivity Analysis

Using the scoring model developed above, WUDBF varied the number of deployments in M3 in order to determine which value maximizes total score. Figure 3.3 below shows how intermediate variables in the scoring model including all-up weight, wing area, GM time, and wingspan varied with the number of deployments in M3.

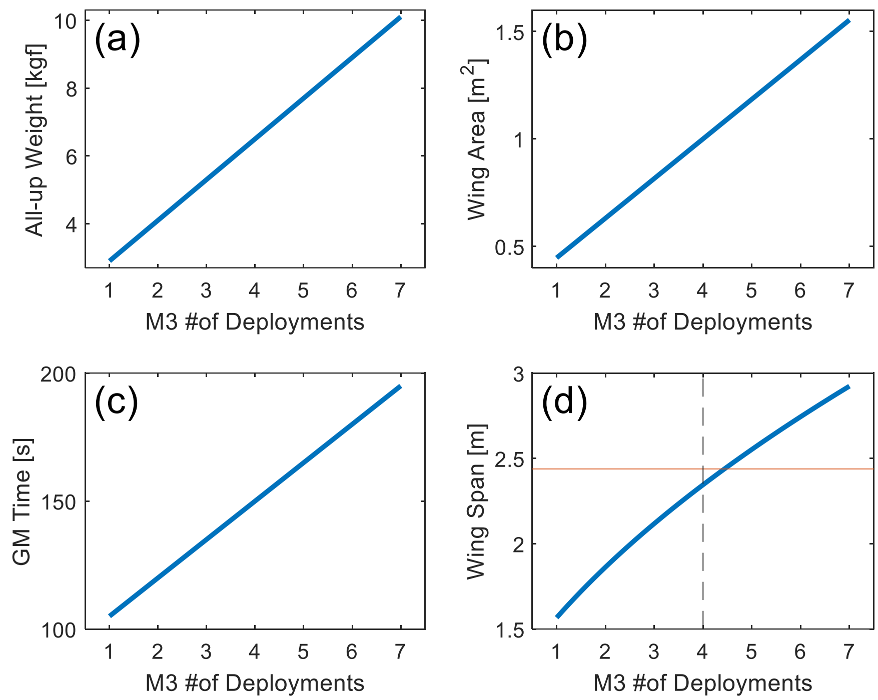


Figure 3.3: Plots of Intermediate Variables in Scoring Model

The plot in Figure 3.3d sets an important constraint on the scoring analysis. Since wingspan cannot exceed 8 ft (2.44 m), based on this model the maximum number of deployments in M3 that can be safely achieved is four. Note that by increasing the assumed wing loading or decreasing the assumed AR, more deployments could be achieved within the 8 ft wingspan constraint. However, WUDBF prioritized a minimal risk design and stuck with the assumed values. The resulting scoring calculations are shown in Figure 3.4 below.

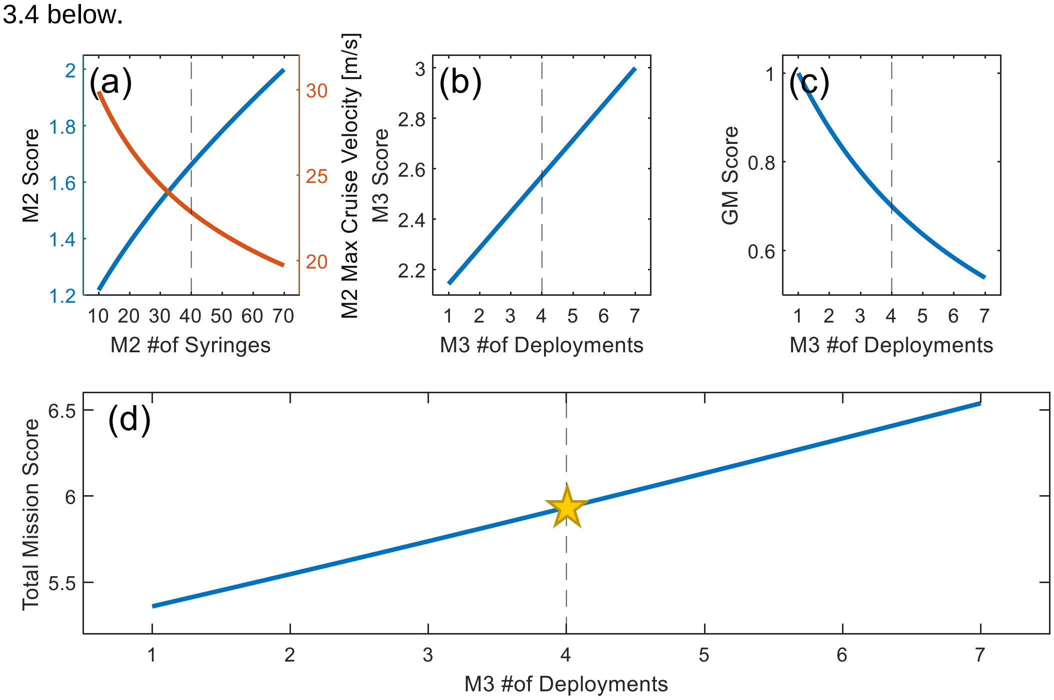


Figure 3.4: Plots of Mission Scores in Scoring Model

For M2, max cruise velocity decreases with increasing number of syringes, which increases lap time. However, the overall M2 score still increases with increasing number of syringes despite the increase in lap time. For M3, score increases linearly with the number of deployments, as expected. For the GM, score decreases with the number of deployments since more time is needed to load the additional payload. Summed together, Figure 3.4d shows that Total Mission Score increases with the number of deployments in M3. Thus, this scoring sensitivity analysis indicates that the optimal design is one that maximizes M2 and M3 payload, while sacrificing cruise velocity and GM time.

Since the maximum number of deployments in M3 is four (based on the wingspan constraint), WUDBF chose four deployments in M3 as the desired design point. From this, the following initial design targets were identified using the plots in Figures 3.3 and 3.4:

Table 3.3: Initial Design Targets

Design Specification	Initial Target Value
Number of deployments in M3	4
Number of syringes in M2	40
Max M2 cruise velocity [m/s]	22.5
GM Time [s]	150
M3 All-up weight [kgf]	6.50
Wing area [m^2]	1
Wingspan [m]	2.35

These design targets were used to guide the initial conceptual design and were further refined in the preliminary design phase.

3.4 Configuration Selection

Once initial design targets were determined, decision matrices were then used to select the overall aircraft configuration as well as the configuration of major subsystems. All decision matrix weights were determined based on design goals and competition requirements. Individual criterion scores were then determined based on team knowledge and experience from previous competitions. The individual criterion scores were multiplied by the weights and summed together to determine the total weighted score for each configuration. A selection was then made by comparing the total weighted score.

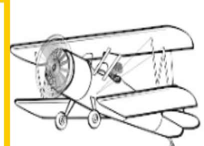
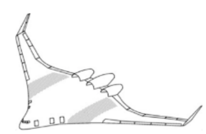
3.4.1 Overall Aircraft Configuration

WUDBF first considered three overall aircraft configurations including the monoplane, biplane, and blended body configurations. Key design criteria included aerodynamic efficiency, stability and control, wing area, and feasibility. The highest weights were assigned to aerodynamic efficiency and stability control, as those are the main drivers of overall aircraft design. Based on these criteria, the following conclusions were made.

- **Monoplane:** Provides good aerodynamic efficiency, stability and control, and is likely the most feasible configuration. However, available wing area within the span limitation is the lowest of the considered configurations.
- **Biplane:** Potentially allows for more lift within the span limitation by maximizing available wing area. However, the increased drag and added structural weight associated with the configuration decreases efficiency.
- **Blended Body:** Allows for increased wing area while still maintaining relatively good aerodynamic efficiency. However, the complex geometries of the blended-body reduces design and manufacturing feasibility. Additionally, the blended body configuration would have limitations in regard to stability and control.

Translating these judgments to the decision matrix resulted in the Table 3.4 below:

Table 3.4: Aircraft Configuration Decision Matrix

		Score		
				
Criteria	Weight	Monoplane	Biplane	Blended Body
Aerodynamic Efficiency	9	5	3	4
Stability and Control	9	5	3	2
Wing Area	5	3	5	4
Feasibility	7	5	3	1
	Total:	140	100	81
	Normalized:	**1.000**	**0.714**	**0.579**

Ultimately, the monoplane configuration was selected due to its superior total weighted score, efficiency, stability, and feasibility. While the monoplane configuration did not offer any advantages with regards to increased wing area, it was thought that the 8 ft wingspan limitation allowed for a large enough wing area to meet mission requirements with a monoplane design.

3.4.2 Wing Placement

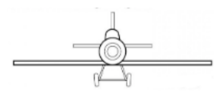
WUDBF next considered wing placement by comparing high-, mid-, and low-wing configurations. Stability and control and takeoff capability were weighted highest followed by feasibility, payload loading, and payload capacity.

- **High-Wing:** Possesses the best stability and control due to its tendency to self-correct. Also is the most feasible configuration since WUDBF used it successfully in the past. However, access to the fuselage from above is restricted, increasing payload loading complexity and GM time.

- **Mid-Wing:** Has stability and control characteristics between that of the low- and high-wing configurations. However, this configuration necessitates that the structural support for the wing runs through the center of the fuselage. This results in a decreased payload capacity and potentially increased payload loading complexity.
- **Low-Wing:** Potentially reduces takeoff length due to ground effects. Because the low-wing configuration allows for easy access from above, it has superior payload loading capability. However, is the least stable configuration due to the CG being above the wing.

The resulting decision matrix is shown in Table 3.5 below:

Table 3.5: Wing Placement Decision Matrix

		Score		
				
Criteria	**Weight**	High-Wing	Mid-Wing	Low-Wing
Stability and Control	9	5	4	3
Takeoff Capability	8	3	4	5
Payload Loading	6	2	3	5
Payload Capacity	6	5	3	5
Feasibility	7	5	4	3
	Total:	146	132	148
	Normalized:	**0.986**	**0.892**	**1.000**

Ultimately, the low-wing configuration was chosen due to its superior takeoff and payload loading capabilities. WUDBF determined that concerns with stability and control could likely be mitigated by adding dihedral to the wing.

3.4.3 Empennage Configuration

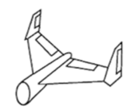
WUDBF then selected the empennage configuration out of the conventional tail, the T-tail, the H-tail, and the V-tail. The main design criteria for empennage configuration were determined to be stability and control, flutter, drag, wake interference during takeoff, and feasibility.

- **Conventional:** Simple to design and manufacture. Also possesses good stability and control and flutter characteristics.
- **T-tail:** Keeps the HT in undisturbed air, which would reduce the drag. However, the high-mounted HT can also generate low flutter speeds and asymmetric lift and yaw which induces high torsional loads. Additionally, having the HT at the tip of the VT introduces additional complexity into the structure and manufacturing process.

- **H-tail:** Offers increased HT efficiency due to the endplates. However, the flutter speed of the H-tail is reduced due to the weight of the VTs being located at the tips of the HT. Additional control and structural complexity is introduced by having two VTs, decreasing feasibility.
- **V-Tail:** Possesses good flutter characteristics and reduces the wake interference at takeoff. However, the V-tail control system is significantly harder to design and implement.

Based on these judgments, the decision matrix in Table 3.6 was constructed.

Table 3.6: Empennage Configuration Decision Matrix

		Score			
					
Criteria	Weight	Conventional	T-Tail	H-Tail	V-Tail
Stability and Control	9	5	4	3	3
Flutter	8	5	3	2	4
Drag	6	4	5	3	2
Wake Interference at T-O	4	3	5	3	4
Feasibility	7	5	3	3	1
	Total:	156	131	94	94
	Normalized:	**1.000**	**0.840**	**0.603**	**0.603**

Ultimately, the conventional tail was chosen because it was considerably more feasible than the other configurations and provided improved stability/control and flutter characteristics with comparable wake interference.

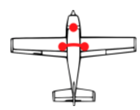
3.4.4 Landing Gear Configuration

The landing gear was then chosen based on ground maneuverability, payload deployment, drag, and structural strength. Using these criteria, the following configurations were compared in Table 3.7.

- **Tricycle:** Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.
- **Bow Tricycle:** One front independent strut and two rear wheels which are connected to the fuselage via a common strut or fairing. This configuration allows good ground maneuverability while also increasing structural strength by adding bowed support to the two rear wheels. However, the structural strength of the front landing gear and takeoff capability are a concern.

- **Strut Tail-dragger:** Two front wheels connected independently to the fuselage and one rear wheel connected to the tail. This configuration increases takeoff capability and potentially eases payload deployment by lowering the rear of the aircraft closer to the ground. However, it provides limited ground maneuverability and offers limited structural strength.
- **Bow Tail-dragger:** Two front wheels connected to the fuselage via a common strut or fairing and one rear wheel connected to the tail. Offers the same improved takeoff capability and payload deployment as the strut tail-dragger, while also providing better structural support. Ground maneuverability of this configuration is limited.

Table 3.7: Landing Gear Configuration Decision Matrix

		Score			
					
Criteria	**Weight**	Tricycle	Bow Tricycle	Strut Taildragger	Bow Taildragger
Takeoff Capability	8	3	3	5	5
Payload Deployment	8	3	3	4	4
Structural Strength	7	3	4	3	4
Ground Maneuverability	6	5	5	3	3
Drag	4	3	3	4	4
Feasibility	7	3	4	3	5
	Total:	132	146	148	169
	Normalized:	**0.781**	**0.864**	**0.876**	**1.000**

Despite poor ground maneuverability, the bow tail-dragger configuration was chosen because of the takeoff advantages offered by the increased initial AoA, the improved payload deployment capability, and the added structural integrity provided by the bowed front gear.

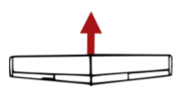
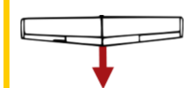
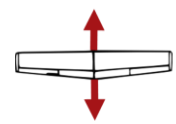
3.4.5 Propulsion Configuration

The propulsion system was chosen based on its efficiency, thrust, weight, component interference, and drag. The following propulsion configurations were considered and the constructed decision matrix is shown below in Table 3.8.

- **Tractor:** Single motor on the front of the aircraft. Provides the highest efficiency, lowest weight and drag, and is the most feasible. Ground clearance could be a concern depending on the required propeller size and height of front landing gear.
- **Pusher:** Single motor aft of the fuselage. Provides low weight and drag but has a decreased ground clearance (especially during rotation) and low efficiency due to fuselage interference.

- **Twin:** Two motors which are mounted on the wings. Provides improved ground clearance due to reduced propeller size and reasonable efficiency. Also offers torque balancing and is reasonably feasible. However, drag is increased resulting from the placement of the motors on the leading edge of the wing. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.
- **Push/Pull:** One motor at the front and aft of the fuselage. Offers low weight and drag. However, is the least efficient due to propwash interference, has the same ground clearance concerns as the tractor and pusher configurations, and is the least feasible. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.

Table 3.8: Propulsion Configuration Decision Matrix

		Score			
					
Criteria	**Weight**	Tractor	Pusher	Twin	Push-Pull
Efficiency	8	5	3	4	2
Weight	8	5	5	4	4
Ground Clearance	7	4	2	5	2
Drag	6	5	5	3	4
Feasibility	4	5	3	4	2
	Total:	158	120	133	94
	Normalized:	**1.000**	**0.759**	**0.842**	**0.595**

The tractor configuration was ultimately chosen because it was thought to maximize efficiency while minimizing weight and drag. Additionally, the tractor configuration is the most feasible design and is one that WUDBF has successfully employed in the past.

3.4.6 M3 Payload Configuration

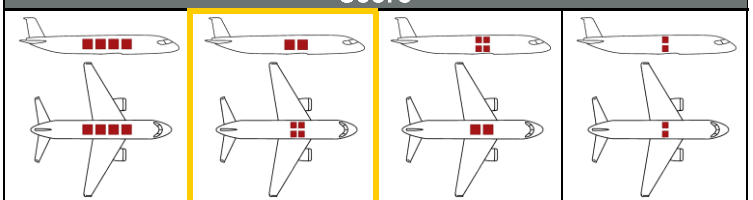
The M3 payload configuration was chosen based on longitudinal weight distribution, lateral weight distribution, vertical weight distribution, ease of payload deployment, and available fuselage space. The following payload arrangements were compared in Table 3.9.

- **4 in-line:** Payloads on one horizontal plane lined up in the longitudinal direction. Minimizes lateral and vertical CG shifts and would make payload deployment reasonably simple. However, this configuration results in the largest longitudinal CG shift and necessitates a long fuselage.
- **Planar 2 by 2:** Payloads on the same plane, arranged in two longitudinal and two lateral rows. This configuration minimizes the vertical CG shift while balancing the longitudinal and lateral CG

shifts. This arrangement also improves payload deployment but necessitates a fuselage wide enough to fit two payloads side-by-side.

- **2 on 2 in-line:** Payloads stacked on two horizontal planes and arranged into one row running longitudinally along the fuselage. This configuration minimizes lateral CG movement while balancing longitudinal and vertical CG shifts. However, this arrangement increases payload deployment complexity due to the stacked payloads.
- **2 on 2 side-by-side:** Payloads arranged on two horizontal planes and two lateral rows to minimize longitudinal CG shifts while balancing vertical and lateral shifts. Like the 2 on 2 in-line, this configuration also increases payload deployment complexity.

Table 3.9: Payload Configuration Decision Matrix



		Score			
Criteria	**Weight**	4 In-Line	Planar 2 by 2	2 on 2 In-Line	2 on 2 Side-by-Side
Longitudinal CG Shift	9	1	3	3	5
Vertical CG Shift	8	5	5	2	2
Lateral CG Shift	6	5	2	5	2
Payload Deployment	6	3	5	2	1
Fuselage Size	4	3	3	4	4
	Total:	109	121	101	95
	Normalized:	**0.901**	**1.000**	**0.835**	**0.785**

In the end, the planar 2 by 2 was chosen because it is equally efficient in both the longitudinal and lateral directions. Thus, it does not require an extraordinarily wide fuselage, and does not induce large CG shifts as the payloads are deployed. Additionally, it does not require more than one horizontal plane of stacking, which simplifies payload deployment.

3.5 Final Conceptual Design

The final conceptual design consists of a low-wing, monoplane, tail-dragger aircraft with a conventional tail and a propulsion system in the tractor configuration. The aircraft is designed to carry an M2 payload of 40 syringes and a M3 payload of 4 packages. The targeted max cruise velocity in M2 is 22.5 m/s .

4. Preliminary Design

During the preliminary design phase, the initial aircraft concept and configuration from the conceptual design phase was further developed. Key aspects of the design including the overall sizing, wing geometry, airfoil, and propulsion package were refined following a number of trade studies.

4.1 Design Methodology

The full design methodology employed by WUDBF is outlined in Figure 4.1 below.

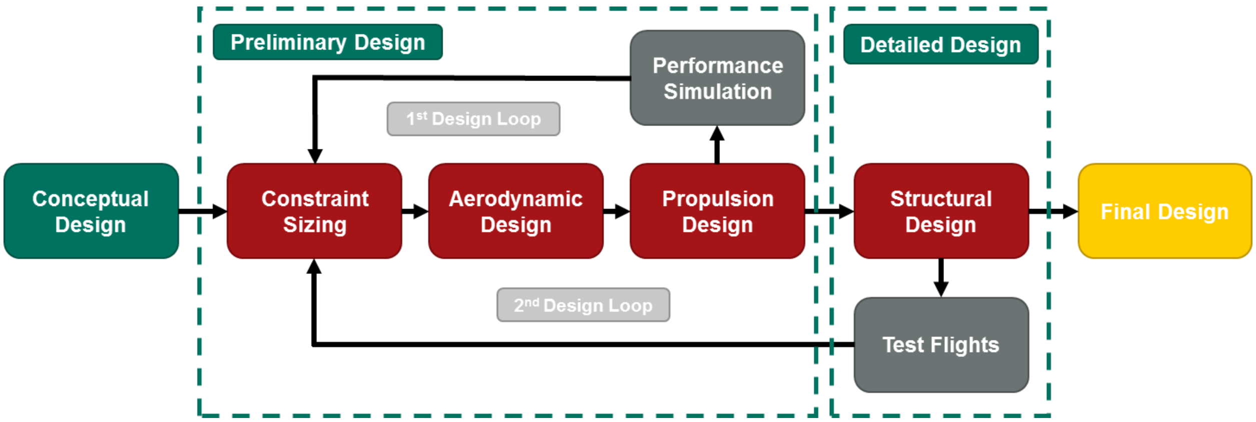


Figure 4.1: WUDBF Design Methodology

For the preliminary design phase, WUDBF utilized an iterative process focused on the performance of the fully loaded aircraft (i.e. with 4 packages in M3). This was done because 1) the AUW for M2 with 40 syringes is expected to be close to that of the fully loaded aircraft in M3 and 2) aircraft are typically designed for the fully loaded case. Initial design targets from the conceptual design phase were used as a starting point. A constraint analysis was performed to size the overall aircraft and to determine the desired wing-loading and required T/W and power. The chosen wing-loading was then used to guide aerodynamic design, from which the airfoil, wing geometry, and tail sizing were determined. The performance and stability characteristics of the aircraft were analyzed using the vortex lattice method in the analysis tool XFLR. The drag polar of the aircraft was determined using induced drag data from XFLR and the component drag build-up method. Following aerodynamic design, propulsion design took place using the required T/W, the required power, estimated AUW, and the drag polar as a starting point. The online tool eCalc was used to sort through a database of components to identify candidate propulsion system combinations. Data from eCalc was then used to select the optimal system, and performance was confirmed using WUDBF's time domain propulsion simulator written in MATLAB.

In order to iterate the design, WUDBF employed two design loops. Following the first design iteration, the preliminary design was tested using KM Sim, a kinematic mission simulator developed by WUDBF. Performance estimates from KM Sim were then used to iterate the preliminary design. Once the design met desired performance goals in KM Sim, the design moved into the second design loop. Structural design took place to create a detailed CAD model of the aircraft. This model was used to build a prototype aircraft. Flight test data from the prototype were then used to inform changes to the preliminary design. Ultimately, the final design developed as the culmination of several iterations, with each iteration improving on performance and mission scores.

4.2 Constraint Sizing

An initial constraint sizing was performed by plotting the required thrust-to-weight ratio (T/W) to achieve a 25 ft takeoff and cruise at 22.5 m/s for a range of wing-loadings (W/S) using the equations developed in Gudmundsson [2, ch. 3]. These plots are shown in Figure 4.2a. Since the aircraft must be propeller driven using an electric motor, it is useful to convert T/W to required electrical power using initial estimates of weight, mechanical efficiency, and propulsive efficiency. This also allows for the setting of an upper power constraint of 2500 W (estimated from previous experience and past competition aircraft). The resulting power required curves are plotted in Figure 4.1b.

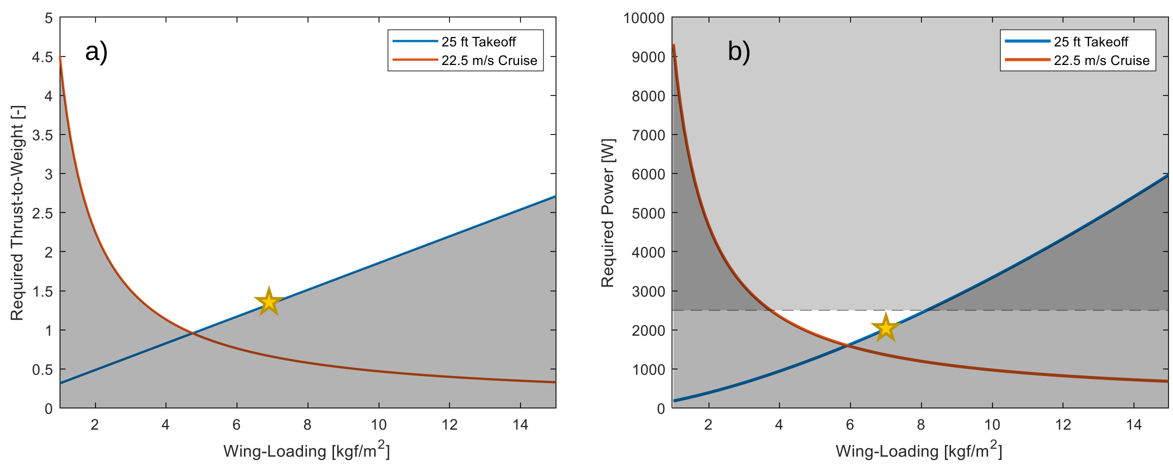


Figure 4.2: Plots of Required T/W and Power vs. Wing-Loading

From Figure 4.2b, the viable design space is limited to a small region of wing-loadings between 3.5 kgf/m^2 and 8 kgf/m^2. Note that the assumed wing-loading of 6.5 kgf/m^2 used for the scoring sensitivity analysis in Section 3.3 falls within this range, further justifying the assumption.

Ultimately, a wing-loading of 7.0 kgf/m^2 was selected for the design point as a conservative choice. During past competitions, the final WUDBF aircraft has consistently exceeded the designed weight. Thus, designing an aircraft with a wing-loading on the higher end of the acceptable range allows for a larger structural weight margin (given that propulsion and payload weight are essentially constant).

From the chosen wing-loading of 7.0 kgf/m^2, the minimum required power was set at ~2100 W and the minimum required T/W ratio at ~1.4 (as indicated by the stars in the figures above). Note that from the plots, the required T/W and power values are determined by the takeoff constraint; the required power at cruise is only ~1400 W. Thus, the propulsion system could be designed to cruise at a lower throttle setting to increase endurance.

4.3 Initial Weight Estimation

The AUW of the aircraft was estimated using a breakdown of individual components. The weight of the wing, HT, VT, and fuselage were estimated using previous aircraft, which were built with similar methods and materials. Battery, motor, servo, electronics, central CF rod, and landing gear weights were determined by weighing representative components already possessed by WUDBF. The weight of the deployment mechanism was estimated using those built by WUDBF for recent competitions. The M3 payload weight is based on that given in the rules document. The full breakdown of estimated weights is shown below in Table 4.1.

Table 4.1: AUW Estimate Breakdown

Component	Weight (kgf)
Wing	2.15
HT	0.20
VT	0.13
Fuselage	0.50
Battery	0.75
Motor	0.61
Servos	0.08
Electronics	0.20
Central CF Rod	0.082
Front Landing Gear	1.0
Rear Landing Gear	0.10
Deployment Mechanism	0.20
M3 Full Payload	0.907
Total:	6.909

Based on this analysis, the predicted AUW of the fully loaded aircraft during M3 is 6.909 kg. Using the wing-loading of 7.0 kgf/m^2 determined in 4.2, that sets the desired wing area at around 0.987 m^2.

4.4 Aerodynamic Performance

The aerodynamic design of the aircraft was iteratively improved to ensure that it is stable, controllable, and achieves maximum aerodynamic performance. XFLR5 and MATLAB were the primary tools used for aircraft aerodynamic analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were dimensioned to improve handling characteristics and minimize takeoff distance, respectively.

4.4.1 Aspect Ratio

Based on the desired wing area determined in 4.3 of 0.987 m^2, the required wingspan for different AR could be plotted as shown in Figure 4.3 below.

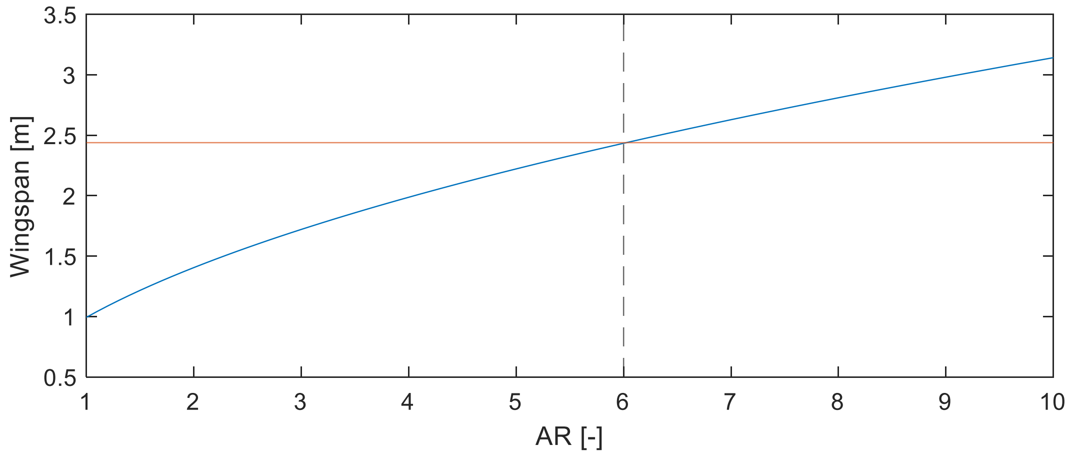


Figure 4.3: Required Wingspan for Varying AR

Based on the maximum 8 ft (2.44 m) wingspan limit, the viable AR ranges from 1-6. Generally, wings with an AR on the lower end of the range have the advantage of increased maneuverability due to increased stall AoA and low roll damping. Lower AR wings also require less structural weight. However, they are inefficient due to high induced drag and are generally more difficult to design and fly. Wings with an AR closer to 6 are more stable with good roll response and limited adverse yaw. They also benefit from improved efficiency due to lower induced drag [2, p. 310]. It is generally agreed that an AR of between 5 and 6 is ideal for an easy flying aircraft [1].

Ultimately, WUDBF chose the highest possible AR of 6 in order to maximize section lift and minimize induced drag. This choice also prioritizes safety, stability, and feasibility over maneuverability, which is in line WUDBF’s stated design requirements.

4.4.2 Main Airfoil Selection

In order to select the main airfoil, the predicted Re of the aircraft was first calculated, using Equation 4.1:

$$Re = \frac{cV_\infty}{v} \tag{4.1}$$

where c is the mean aerodynamic chord [m], V_∞ is the freestream velocity [m/s], and v is the kinematic viscosity of air [m²/s]. Kinematic viscosity was calculated from the average temperature in April (14 ℃) and altitude (400 m) in Wichita which yielded a kinematic viscosity of 1.498x10^{-5} m^2/s. These conditions yielded a Reynold’s number of 626,384.

Airfoil selection was then performed by selecting several different airfoils with maximum C_L/C_D near a Reynold’s number of 500,000. Additionally, the search was restricted to airfoils with a minimum thickness of 10% and a maximum camber of 8% to ensure that the airfoil could accommodate structural and system integration and be reasonably manufactured with the processes available. Five promising airfoils were selected for comparison as shown in the table below:

Table 4.2: Airfoil Selection Options [3]

	Max. C_L/C_D at Re = 500 000	Max. Thickness [% of chord]	Max. Camber [% of chord]
Clark K	102.7	11.7	3.3
Clark Y	98.7	11.7	3.4
NACA 1412	73.5	12	1
NACA 2412	87.3	12	2
USA-35B	106.0	11.6	4.1

Once these potential airfoils were identified, analysis of each airfoil was performed in XFoil and XFLR to determine which provided a favorable lift coefficient, lift-to-drag ratio, and drag bucket at a Reynold's number of 626,000.

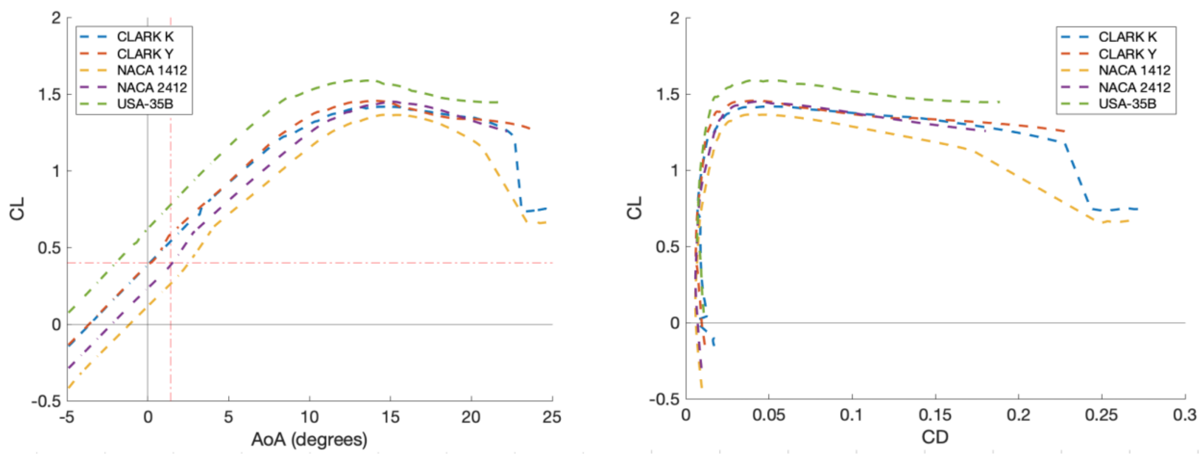


Figure 4.4: Airfoil C_L vs AoA (left) and C_L vs C_D (right)

As shown in the Figure above, the USA-35B possessed the highest lift coefficients and widest drag bucket. While the lift generated from this airfoil would improve T-O performance, it would also produce excessive lift at the target cruise velocity of 22.5 m/s, meaning the aircraft would have to fly slower resulting in longer lap times. Similarly, the Clark Y airfoil would also generate more lift than necessary at the target cruise velocity. While the NACA 2412 did not have the highest lift coefficients, it was found to provide sufficient lift at T-O and appropriate lift to meet cruise requirements. The NACA 2412 also had a drag bucket which comparable to the Clark Y. Thus, the NACA 2412 airfoil was chosen because it had the lowest C_{Dmin}, a wide drag bucket, and the desired lift characteristics at the target cruise velocity.

4.4.3 Empennage Sizing

Stabilizer airfoils are generally symmetric as they may either provide upforce or downforce, depending on the final trim state of the aircraft. WUDBF selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects [4]. Initial sizing of the empennage for stable flight began by investigating the vertical tail (VT) and horizontal tail (HT) volumes for aircraft of the same class. For a single-engine aircraft, the initial estimate was set at around V_{HT} = 0.70 and V_{VT} = 0.04 [5]. Initial horizontal and vertical tail areas were then solved for using Equation 4.2 and 4.3 as shown below [2, p. 513]:

$$S_{HT} = \frac{V_{HT} * S_{ref} * c_{ref}}{I_T} \tag{4.2}$$

$$S_{VT} = \frac{V_{VT} * S_{ref} * b_{ref}}{I_T} \tag{4.3}$$

Where S_{HT} is the horizontal tail area [cm^2], V_{HT} is the horizontal tail volume [cm^3], S_{VT} is the vertical tail area [cm^2], V_{VT} is the vertical tail volume (in cm^3), S_{ref} is the main wing area [cm^2], c_{ref} is the mean geometric chord [cm], b_{ref} is the span of the main wing [cm], and I_T is the optimal tail arm length [cm]. Based on these calculations, the VT and HT planform areas were set at 1285.4 cm^2 and 2236 cm^2 respectively. An initial assumption was made that the AR of the HT would be equal to about half the AR of the main wing and that the AR of the VT would be about 2 [6]. Initial estimates for the span of the VT and HT were then made using Eq 4.4 and 4.5 as shown below [2, p. 513]:

$$b_{HT} = \sqrt{AR_{HT} * S_{HT}} \tag{4.4}$$

$$b_{VT} = \sqrt{AR_{VT} * S_{VT}} \tag{4.5}$$

Where b_{HT} is the horizontal tail span (in cm), and b_{VT} is the vertical tail span (in cm).

During the design process, however, it was determined that using the method outlined by Gudmundsson was not feasible as optimal tail arm length is determined by considering the fuselage as a frustrum. Since it was decided that the fuselage would not extend the length of the tail early on in the design phase to save weight, the frustrum approximation was not reasonable for the design (see figure below). Thus, initial estimates for the tail arm using Gudmundsson's method were found to be sub-optimal.

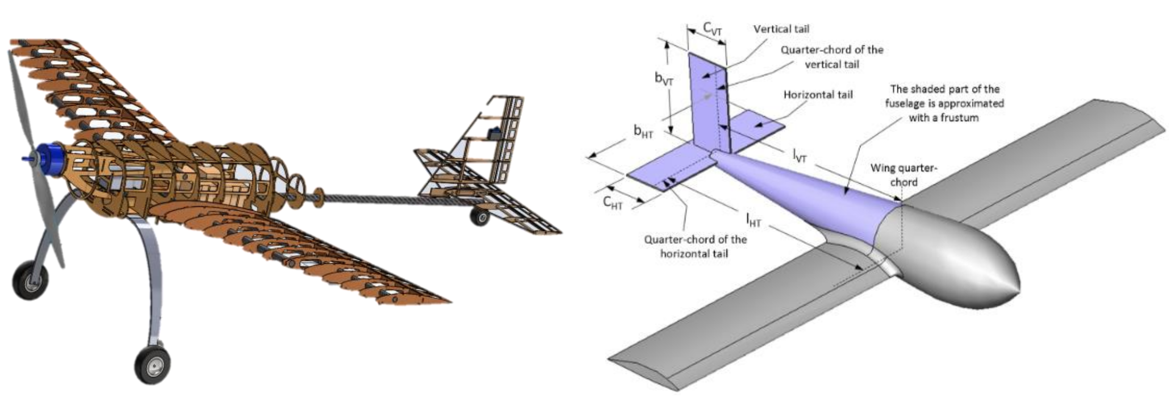


Figure 4.5: Initial Configuration of the Pflyzer (left) vs. Simplified Aircraft Layout used by Gudmundsson (right) [2].

To correct for this, the tail arm, CG locations and horizontal and vertical tail areas determined above using Gudmundsson's method were iterated on throughout the entire design process based on dynamic stability analysis and static margin. In the end, the resulting VT had a planform area of 1209.5 cm² and the HT had an area of 1845.0 cm².

4.4.4 High Lift Devices

Winglets were implemented to increase lift performance and were sized according to Gudmundsson and previous testing done by WUDBF [4]. Tests concluded that the additional lift generated as a function of winglet area hits a plateau beyond a certain point. Since a 5° dihedral angle was added to the wings, ground clearance was not a concern in this design. This allowed the area of the winglet to be larger than it would have been otherwise. The final winglet area was set at 230 cm². The Manufacturing and Structures subteams determined that this design is feasible and the additional lift generated outweighs the increased manufacturing complexity.

4.4.5 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using rule-of-thumb parameters from Gudmundsson [2, p. 419]. Control surface sizing was verified using Equation 4.6 (see below) from Gudmundsson [2, p. 953], previous WUDBF aircraft, and prototype performance.

$$C_{l_{\delta_a}} = \frac{c_{l_{\delta_a}} C_R}{Sb}\left[(b_2^2 - b_1^2) + \frac{4(\lambda - 1)}{3b}(b_2^3 - b_1^3)\right] \tag{4.6}$$

Where $C_{l_{\delta_a}}$ is the aileron authority, $c_{l_{\delta_a}}$ is the change in lift coefficient with aileron deflection, b is the wing span [m], S is the wing area [m^2], C_R is the root chord [m], and λ is the taper ratio.

Plain flaps were chosen to simplify the manufacturing process. The hinge point of the flaps was placed at the center of the leading edge arc to minimize large gaps and overhangs when the surfaces are deployed. The flaps were given a chord fraction of 25%, and a span of 40% of the wing. The ailerons had a chord fraction of 25% and a span of 60% of the wing. Both the flaps and the ailerons were designed to deploy at a 25° angle. Using data taken from XFLR, Figure 4.5 shows the plotted lift coefficient versus AoA with the flaps deployed at 25° and with no flaps. Chord fractions for the control surfaces on the empennage were 30% for the elevator, and 40% for the rudder. The Rudder was designed to deflect a maximum of 25°, while the elevator was designed to deflect a maximum of 25°.

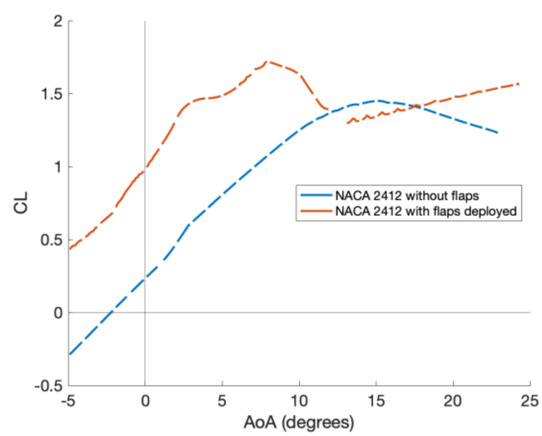
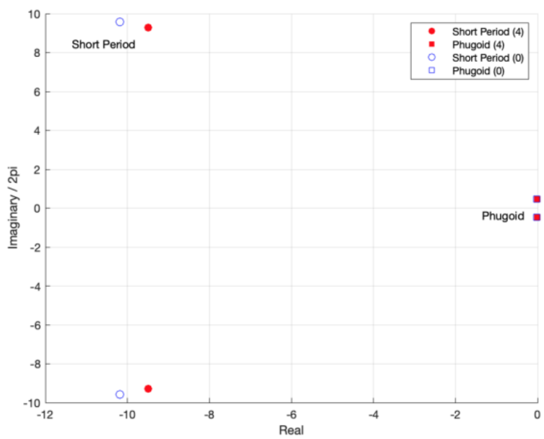


Figure 4.6: C_L vs AoA With and Without Flaps

4.4.6 Dynamic Stability Analysis

The dynamic stability characteristics of the aircraft were analyzed using XFLR. The Root Locus plot below in Fig 4.7 shows the dynamic stability modes when the aircraft is loaded with four packages in M3 (red) and when it is empty (blue). Based on the weight of the packages, the stability when loaded with four packages in M3 is nearly identical to M2 stability when loaded with syringes. Thus, the dynamic stability characteristics of the M3 fully-loaded configuration were comparable to M2, so M2 is not shown. As packages are deployed in M3, the CG shifts, which negatively impacts dynamic stability. However, the dynamic stability characteristics with 1, 2, and 3 packages are still acceptable, with values between those for the fully loaded and empty aircraft. Thus, the 1, 2, and 3 package configurations are also not shown.



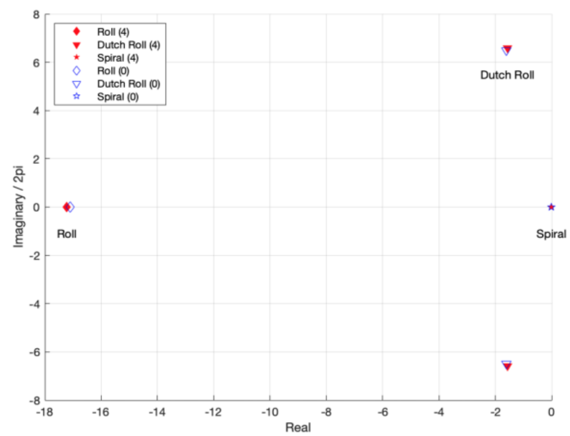


Figure 4.7: M3 Root Locus Longitudinal plot (left) and Lateral plot (right)

Based on the root locus plots, all modes were stable for both loading conditions. The design goal of this aircraft was to satisfy the level 1 flying qualities of a Class 1 aircraft described in MIL-F-8785C [7]. The dynamic stability characteristics from XLFR when the aircraft was loaded with 4 packages are shown below in Table 4.3 with MIL-F-8785C requirements in parenthesis. Similarly, the stability characteristics of the aircraft when empty are shown below in Table 4.4.

Table 4.3: Dynamic Stability Table for M3 Fully Loaded

Mode	ζ	ω_n [rad/s]	$\zeta\omega_n$ [rad/s]	τ [s]
Short Period	0.714 ($0.35 < \zeta < 2.00$)	13.2889	9.4883	-
Phugoid	0.028 ($\zeta > 0.04$)	0.4712	0.0132	-
Roll	-	-	-	0.058 ($\tau < 1.4$)
Dutch Roll	0.232 ($\zeta > 0.19$)	6.7669 (>1.0)	1.5699 (>0.35)	-
Spiral	-	-	-	119.728

Table 4.4: Dynamic Stability Table for Empty Aircraft

Mode	ζ	ω_n [rad/s]	$\zeta\omega_n$ [rad/s]	τ [s]
Short Period	0.728 (0.35 < ζ< 2.00)	13.9864	9.4883	-
Phugoid	0.032 (ζ>0.04)	0.4775	0.0153	-
Roll	-	-	-	0.058 (τ<1.4)
Dutch Roll	0.239 (ζ > 0.19)	6.7669 (>1.0)	1.6173 (>0.35)	-
Spiral	-	-	-	82.198

With the exception of the phugoid damping ratio and the short period natural frequency, all modes satisfied the MIL-F-8785C level 1 requirements. The phugoid damping ratio did satisfy MIL-F-8785C level 2 requirements and pilot evaluation during test flights. The short period frequency fell just outside of level 2, but satisfied pilot evaluation during test flights.

4.4.7 Drag Analysis

Total aircraft drag was calculated using the *component drag build-up method* (CDBM) as described by Gudmundsson [4]. The method models the total drag C_D as the sum of the minimum drag C_{Dmin} and the lift induced drag $C_{D,i}$. The minimum drag coefficient, C_{Dmin}, AKA the zero-lift or parasitic drag coefficient was calculated using the following expression:

$$C_{Dmin} = \left(\frac{1}{S_{ref}}\sum_{i=1}^{N} C_{f_i} \times FF_i \times IF_i \times S_{wet_i} + C_{D_{misc}}\right) \times \left(1 + \frac{CRUD}{100}\right) \tag{4.7}$$

Where S_{ref} is the reference area (in cm^2), C_f is the skin friction coefficient, FF is the form factor, IF is the interference factor, S_{wet} is the wetted area of the component (in cm^2), and $C_{D_{misc}}$ is the drag coefficient from miscellaneous components other than the fuselage, HT, VT, and main wing.

The skin friction coefficient for the main wing was determined using Young's method for mixed laminar-turbulent flow [2, p. 679]. The FF for the main wing was found using Raymer's method with the compressibility correction term set equal to one [5]. Finally, the IF was found by using a typical IF for an un-filleted low wing. The skin friction coefficients for the HT and VT were also determined using Young's method for mixed laminar-turbulent flow, and the FFs were found using Jenkinson's method [2, p. 701]. A typical IF for a conventional tail was used. The skin friction coefficient for the fuselage was determined by considering a 100% laminar boundary layer. The FF of the fuselage was then estimated using the method suggested by Hoerner [8, p. 6-17] for bodies in airflow Re > 10^5. Air inlet vents and winglet endplates were accounted for in the C_{Dmisc} term using equivalent parasitic coefficients. To ensure that the drag estimate remained conservative, drag reduction from winglet endplates was ignored. To include drag contributions that could not otherwise be accounted for, a CRUD factor of 25% was applied as recommended by Gudmundsson (see Equation 4.7). From these calculations, the minimum drag coefficient was found to be 0.1553. The full parasitic drag breakdown is presented below:

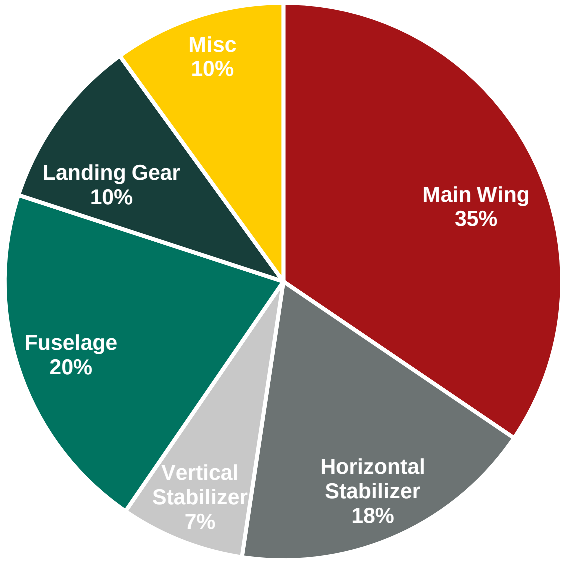


Figure 4.8: Parasitic Drag Breakdown

Table 4.5: Parasitic Drag Breakdown

Component	$C_{Dmin,i}$
Main Wing	0.04280
Horizontal Stabilizer	0.02220
Vertical Stabilizer	0.00894
Fuselage	0.02536
Landing Gear	0.01240
Misc	0.01239
C_{Dmin} w/o CRUD	0.12410
C_{Dmin} w/ CRUD	**0.1553**

As expected, the fuselage and main wing were the largest contributors to parasitic drag. The parasitic drag was then added to the lift induced drag, calculated using XFLR, to give the total drag polar, as shown in the figure below.

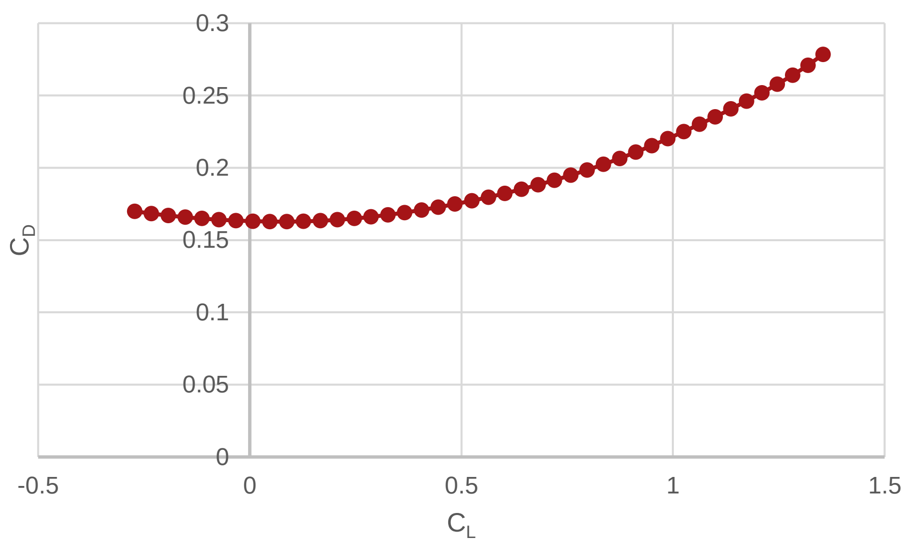


Figure 4.9: Total C_D vs. C_L

4.5 Propulsion

Design of the propulsion system centered on meeting a number of requirements set earlier on in the design process. In particular, a required T/W of ~1.4 for T-O, required electrical power of ~2100 W and ~1400 W for T-O and cruise, respectively (see Section 4.2), a desired cruise velocity of 22.5 m/s (see Section 3.3), and enough endurance to complete four laps (and four takeoffs) in M3. Additional considerations included maximizing efficiency, especially during cruise, and minimizing overall propulsion system weight.

4.5.1 Battery Selection

The DBF rules allow for the use of either LiPo (Lithium Polymer) or NiMH (Nickel-Metal-Hydride) batteries for propulsion. While NiMH batteries tend to be more robust and safe, LiPo batteries offer higher energy density, power to weight ratio, discharge rate, efficiency, and reliability. LiPos also tend to take up less space for a given energy capacity. Thus, WUDBF chose to use LiPo batteries for the propulsion pack.

LiPo's are commonly available in 3s, 4s, 5s, 6s, and 8s packs. For an aircraft of this size, LiPos below 6s are unlikely to provide the performance required. For 8s LiPos, there are not many (if any at all) available below the 100 Wh (3378.38 mAh for 8s) limit. Thus, WUDBF chose to use 6s LiPos, since they are readily available below the 100 Wh (4504.50 mAh for 6s) limit and have been used in the past to power competition aircraft of a similar size to this year's design.

In order to size the battery, WUDBF focused on M3, the mission that requires the greatest endurance. WUDBF plans to perform 4 deployments, meaning four laps and four takeoffs. Based on a desired cruise velocity of 22.5 m/s and a lap distance of 762 m (2500 ft), the aircraft must be able to fly for 2.26 minutes during M3. In order to account for takeoffs and to add a factor of safety, WUDBF set the desired M3 endurance at 3.5 minutes. Based on the earlier constraint analysis, the estimated power required at cruise is ~1400 W and at takeoff is ~2100 W. WUDBF estimated that the plane will spend approximately 5% of a lap under "takeoff" conditions and 95% under "cruise" conditions, yielding a weighted power requirement of 1435 W. Flying for 3.5 minutes with a 1435 W power consumption gives a required energy of 301.35 kJ. For a 6s LiPo (22.2 V nominal), assuming 85% max discharge, the required capacity is therefore 4436.05 mAh. See equation below for full calculation.

$$3.5\,min \times \frac{60\,s}{1\,min} \times 1435\,\frac{J}{s} \div 22.2\,\frac{J}{C} \times \frac{1\,mAh}{3.6\,C} \div 0.85 = 4436.05\,mAh \tag{4.8}$$

Based on this calculation, WUDBF chose to use a 6s LiPo with a capacity of 4500 mAh, which is just below the 100 Wh limit. The Turnigy Nano-Tech 4500 mAh 6s LiPo battery was selected to serve as the propulsion pack since it met the capacity requirement. Additionally, the Nano-Tech battery also possessed high discharge rates of 45C-Constant and 90C-Burst, which allows for a stronger throttle punch and improved takeoff performance.

4.5.2 Motor and Propeller Selection

In order to identify possible motor-propeller combinations, WUDBF employed the online propulsion design tool eCalc [9]. Using the parameters listed in the table below, WUDBF searched through the eCalc database for motor-propeller combinations that met the stated performance requirements.

Table 4.6: eCalc Search Parameters

Parameter	Value	Reasoning
Max Power [W]	~2100	See Section 4.2
T/W	~1.4	See Section 4.2
Max Cruise Velocity [m/s]	22.5	See Section 3.3
Endurance [min]	3.5	See Section 4.5.1
Battery	6s, 4500 mAh	See Section 4.5.1
Max Prop Diameter [in]	24	Estimated max ground clearance

From this search, a list of over 50 unique motor-propeller combinations were identified. This list was then further refined by limiting the possible motors to only those from trusted manufacturers (based on past experience and general online consensus) that were available, in-stock, at the time of the search. This narrowed the possibilities down to the four motor-propeller combinations shown in Table 4.7:

Table 4.7: Possible Motor-Propeller Combinations

Motor	Prop	Elec. Power @ Max [W]	Current @ Max [A]	RPM (Static)	Static Thrust [kgf]	T/W	Torque [N·m]	Flight Time [min]	Drive Weight [kgf]
SunnySky X7025-30cc-270	24x12	2258.5	112.94	4836	10.821	1.52	4	4.3	1.627
Scorpion SII-4035-330	22x13	2304.6	115.53	5101	10.042	1.45	3.4	4.1	1.439
Hacker A60-5S V4 28-Pole	23x10	2083.7	103.22	5280	9.460	1.33	3.35	4.3	1.615
Hacker A60-7XS V4 28-Pole	22x11	2204.3	109.9	5461	9.742	1.4	3.3	4.2	1.462

All four choices met or nearly met all of the stated requirements. Thus, in order to further evaluate each combination, plots of electrical efficiency vs. throttle, endurance vs. throttle, and thrust vs. electrical power were generated using data from eCalc:

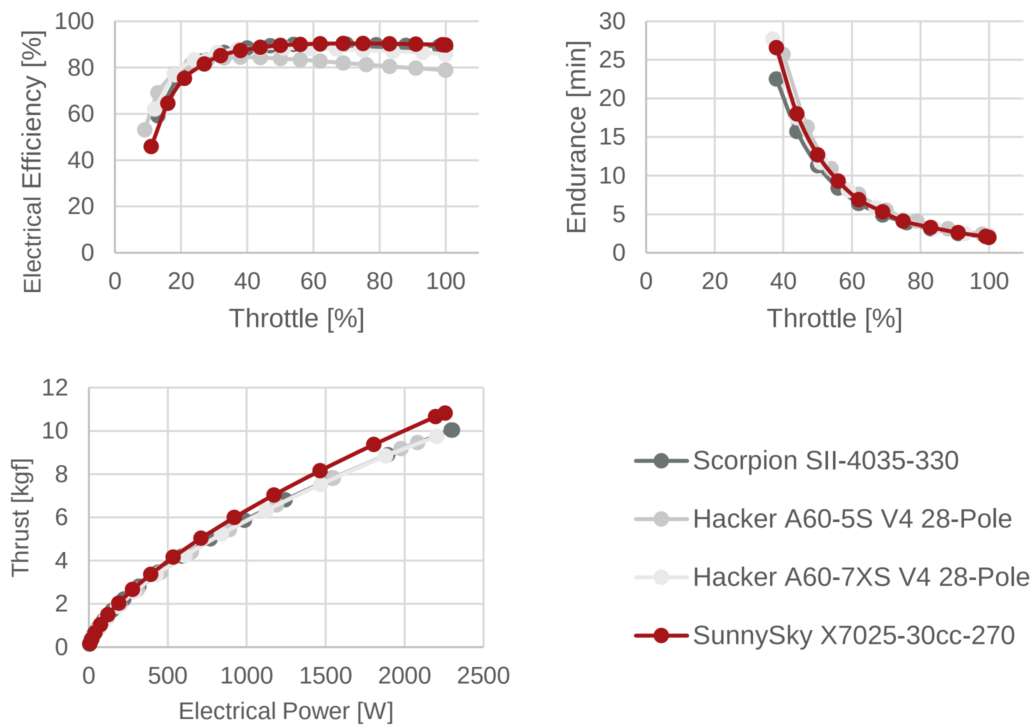


Figure 4.10: Performance Data for Motor-Propeller Combinations

From Figure 4.10, the SunnySky X7025-30cc-270 motor with a 24x12 propeller had the greatest electrical efficiency over relevant throttle settings (60-100%). This motor-propeller combination also provided superior thrust-to-power performance and endurance that was comparable to the other choices. From Table 4.7, the SunnySky did have the highest drive weight, but it still provided the greatest T/W. Based on these factors, the SunnySky X7025-30cc-270 motor with a 24x12 propeller was ultimately chosen for the aircraft's propulsion system. The top right plot in Figure 4.10 shows that an endurance of 5 minutes can be achieved at 70% throttle. Since M3 scoring does not depend on time, and the 10 minute flight window is more than enough to achieve the desired four scoring laps, 70% throttle will be used during M3 to ensure adequate endurance.

4.5.3 ESC Selection

Given the predicted max current of 112.94 A for the chosen propulsion system (see Table 4.7), WUDBF chose to use the AeroStar Advance 150A HV (6~12S) as the ESC. The AeroStar can handle 150 A continuous and 180 A burst current, which provides an adequate safe margin. Additionally, the AeroStar possesses a large aluminum heat sink case for optimal cooling.

4.5.4 Refinement for M2

Given that the propulsion system was designed primarily with M3 in mind, WUDBF next considered whether it could be refined to improve M2 performance (since the competition rules allow for the propeller and battery to be swapped out for different missions).

Starting with the battery, WUDBF considered whether the propulsion pack could be downsized for the M2 to save weight and increase performance. While four laps are required for M3 (to complete WUDBF's goal of four deployments), M2 only requires three laps with a single takeoff. Using a modified version of the calculation described in 4.5.1, WUDBF estimated that the required capacity to complete M2 was 2534.89 mAh. This allows for the use of the Turnigy Graphene Panther 3000mAh 6S 75C battery for M2, saving approximately 0.25 kgf of weight over the Turnigy NanoTech selected for use in M3.

WUDBF next considered whether the propeller could be optimized to increase cruise velocity during M2. In general, increasing propeller pitch increases pitch speed, which increases efficiency at higher velocities and allows for higher cruise velocity. This comes with the sacrifice of efficiency at low airspeeds, negatively impacting static thrust and takeoff performance [5]. In part because there are not many choices for commercially available propellers above a diameter of 22 in suitable for electric aircraft, WUDBF could not find a propeller that increased cruise velocity without decreasing the static T/W and/or endurance below acceptable levels. Thus, the 24x12 propeller was selected for use in both M2 and M3.

4.5.5 Required vs. Available Thrust Analysis

The dynamic performance of the propulsion system was evaluated to verify eCalc's predicted performance and to determine the aircraft's speed envelope. eCalc does not output dynamic thrust data, so this data was instead obtained from the APC Propellers website (the chosen propeller manufacturer) [10]. The APC performance data was generated based on vortex theory using actual propeller geometry and the NASA Transonic Airfoil Analysis Computer Program to estimate section lift and drag. Predicted RPM values from eCalc were used to query the APC dataset at velocities up to 30 m/s, with the corresponding dynamic thrust values calculated using an interpolation script written in MATLAB. The resulting dynamic thrust curve was then plotted along with the required thrust curve (calculated based on the drag polar in Figure 4.9) as shown below:

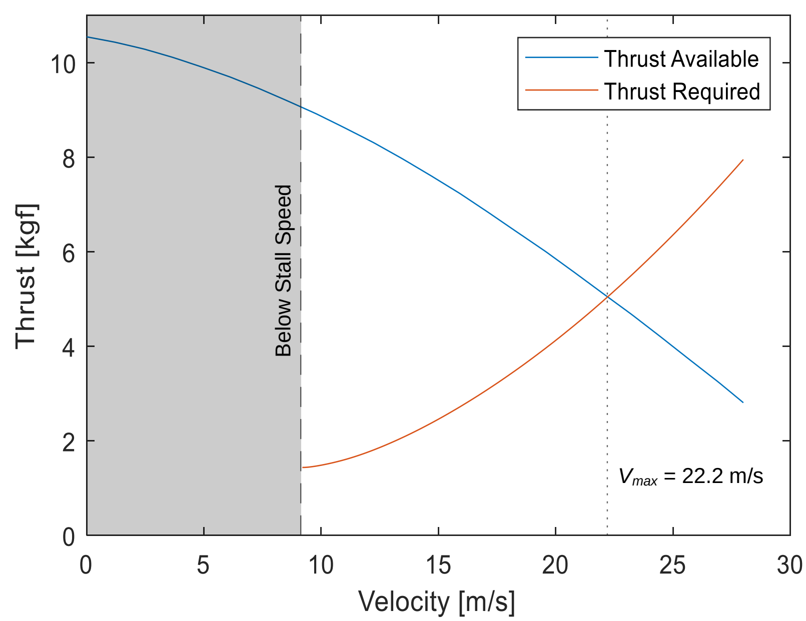


Figure 4.11: Required vs. Available Thrust Curves for Fully Loaded Aircraft

Based on this analysis, the estimated max cruise velocity for the fully loaded aircraft (in both M2 and M3) is 22.2 m/s. This is very close to the designed max cruise velocity of 22.5 m/s. The static thrust predicted using the APC data is 10.54 kgf, giving a T/W ratio of 1.51. This agrees well with both the desired static thrust and the static thrust predicted by eCalc, again confirming that the propulsion system meets the stated performance goals.

4.6 Predicted Aircraft Performance

In order to integrate the various aspects of the preliminary design and predict overall aircraft performance, WUDBF utilized Km_sim, an in-house kinematic mission simulator written in MATLAB [11].

4.6.1 Brief Explanation of Km_sim

Km_sim works by simulating an aircraft's performance as it flies through a mission profile. Each mission profile consists of a series of functions based on simple kinematics and Newton's 2nd Law describing the motion of the aircraft during a particular phase of flight (e.g., takeoff, climb, cruise, etc.). Specific mission profiles were written for M1, M2, and M3. Aerodynamic and propulsion data from Sections 4.4 and 4.5, respectively, were used to generate aerodynamic and propulsion performance look-up tables. These tables allow the simulation to determine the lift, drag, and thrust of the aircraft at any particular time point. During a simulation, Km_sim steps through each mission phase, numerically integrating the relevant equations of motion using ode45 and the look-up tables, while keeping track of changes in the state variables. At the end of the simulation, plots are generated displaying how the state variables change over time as shown in the figures below. For a complete explanation of Km_sim and the source code see [11].

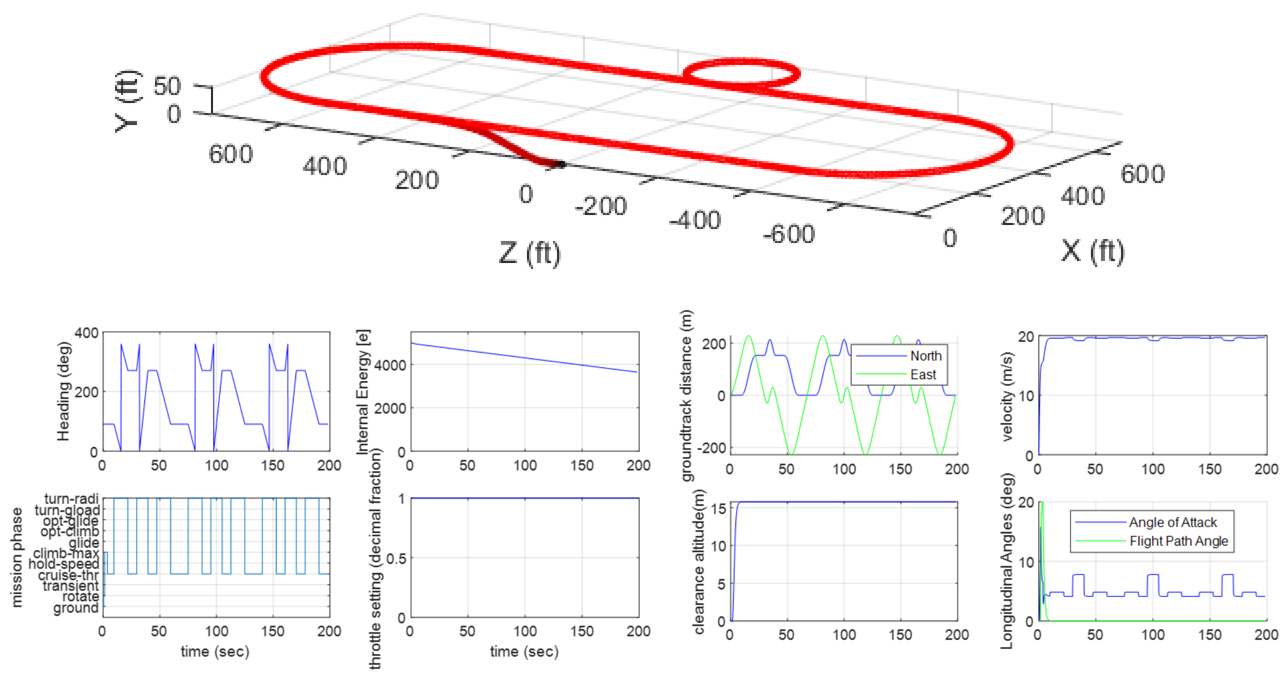


Figure 4.12: Example outputs from Km_sim including a trajectory plot (TOP) and an eight-panel results summary (BOTTOM) showing how state variables changes over time.

4.6.2 Km_sim Results

From simulations of the preliminary aircraft in Km_sim, WUDBF extracted the following aircraft performance predictions. To obtain normalized scores, best scores for M2 and M3 were estimated based on the analysis in Section 3.3.

Table 4.8: Predicted Aircraft Performance

	Mission 1	Mission 2	Mission 3			
Inputs						
All-Up Weight [kgf]	6.020	6.757	6.9072	6.7004	6.4736	6.2468
Payload Weight [kgf]	-	0.7370	0.9072	0.6804	0.4536	0.2268
Wing Loading [kgf/m^2]	6.044	6.784	6.935	6.727	6.500	6.272
$C_{L,max}$	1.44	1.44	1.44			
Static Thrust [kgf]	10.54	10.54	10.54			
Cruise Throttle	1	1	0.7			
Turn Load Factor	2.5	2.5	2.5			
Outputs						
Ground Roll [m]	3.364 (11.04 ft)	3.417 (11.21 ft)	3.511 (11.52 ft)	3.508 (11.51 ft)	3.502 (11.49 ft)	3.499 (11.48 ft)
$C_{L,cruise}$	0.409	0.409	0.394	0.394	0.394	0.394
$C_{D,cruise}$	0.1559	0.1559	0.1566	0.1566	0.1566	0.1566
Cruise Velocity [m/s]	21.68	21.67	20.36	20.38	20.38	20.39
Number of Laps	3	3	4			
Lap Time [s]	48.51	43.97	57.85	57.54	57.53	56.88
Mission Time [min]	2.426	2.198	3.830 (excluding ground phase)			
Scoring						
WUDBF Score	1	18.20	4			
Normalized Score	1	1.614	2.667			

4.7 Uncertainties

The design tools and analysis methods used throughout the preliminary design process all possessed a certain level of uncertainty. The lift and induced drag of the aircraft was computed in XFLR using the vortex lattice method (VLM). Using VLM, lifting surface thickness and air viscosity were neglected. Additionally, the fuselage was excluded from the analysis. Based on previous WUDBF designs, however, it was determined that the decreased lift due to neglecting viscosity is approximately equal to the lift gained by excluding the fuselage. It has also been verified that the contribution of drag associated with wing thickness was small compared to total drag calculations. Therefore, the lift and induced drag of the actual aircraft was expected to be approximately equal to the XFLR predicted lift using the VLM. The parasitic drag estimate using Gudmundsson's method were also subject to a certain level of uncertainty, but that uncertainty was mostly accounted for by the 25% CRUD factor.

eCalc and the APC propeller performance database were the two primary tools used for propulsion design. These third-party tools both carry a certain level of uncertainty owing to the fact that it is not entirely clear how they calculate their outputs. However, this uncertainty is mitigated by the fact that this method of propulsion design has been used successfully by both WUDBF and other DBF teams in the past with an acceptable level of accuracy.

Finally, Km_sim, while a very useful tool, makes a number of assumptions in its performance calculations. For instance, the simulation assumes that the pilot flies the plane on a perfect trajectory. The effects of wind are also neglected in the simulation. Since Km_sim relied on data from the aerodynamic and propulsion designs, any uncertainties from those processes were carried over into the simulation.

Ultimately, the nature of the iterative process meant that the uncertainty associated with any particular design tool was mitigated by ensuring that every aspect of the design agreed with the rest.

5. Detail Design

Following completion of the preliminary design, the structure and external geometry of the aircraft were finalized during the detail design stage. The electronics and package deployment mechanism were also developed during this process. Throughout, structural strength, reliability, weight, and mission performance were considered to refine the competition aircraft.

5.1 Dimensional Parameters

A CAD model of the final aircraft is shown in the figure below. Table 5.1 summarizes the key dimensions of the final aircraft design.

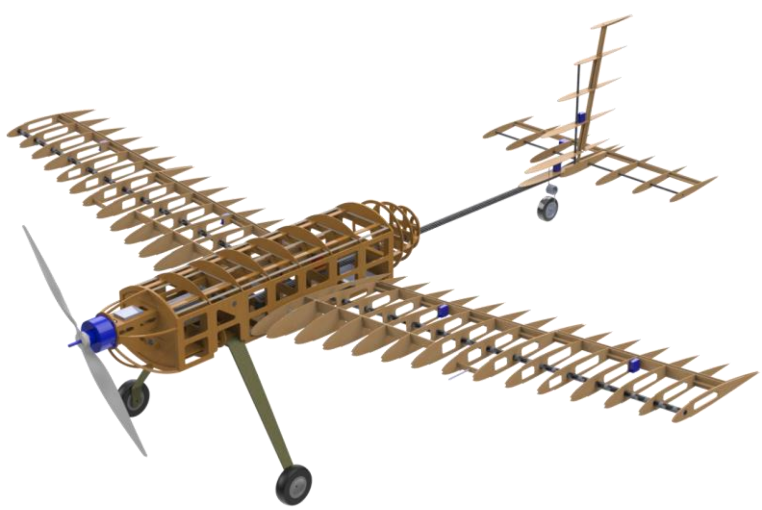


Figure 5.1: CAD Model of Final Aircraft Design

Table 5.1: Key Dimensions of Final Aircraft Design

Vertical Stabilizer		Horizontal Stabilizer	
Span [mm]	820.0	Span [mm]	700.0
Mean Chord [mm]	294.0	Mean Chord [mm]	264.0
Area [m^2]	0.121	Area [m^2]	0.185
AR	2.78	AR	2.64
Wing		**Overall Aircraft**	
Span [mm]	2402.5	Length [mm]	2051.7
Mean Chord [mm]	414.0	Width [mm]	2402.5
Area [m^2]	0.996	Prop Clearance [mm]	165.0
AR	5.78	Initial AoA	11.02°
Dihedral Angle	5°	Empty Weight [kgf]	5.68

5.2 Structural Characteristics and Design Methodology

WUDBF's primary structural design goals were to maximize strength and stability, maximize accessibility to the electronics and payload compartments, minimize weight, and simplify the manufacturing process. The overall structure is designed to direct loads from each subsystem into the major load bearing components of the aircraft. The fuselage consists of wooden formers with a central CF rod running through each former to resist bending moments. The fuselage structure is bolted into the central rod at key points to provide resistance to torsional loads. Wooden side panels also provide torsional resistance and structural stiffness. Propulsive loads from motor thrust and torque that can cause bending, torsion, and vibrations are directed through a wooden motor mount into the fuselage. Ground loads from the landing gear are directed through the fuselage structure and into the central CF rod. The wing consists of wooden ribs connected by stringer and spars, which provide bending and torsional stiffness. Aerodynamic loads from the wing are transferred to the side panels, where they are then distributed across the fuselage structure and the central CF rod. The empennage is directly connected to the central CF rod via CF spars, distributing loads into the central rod and the fuselage.

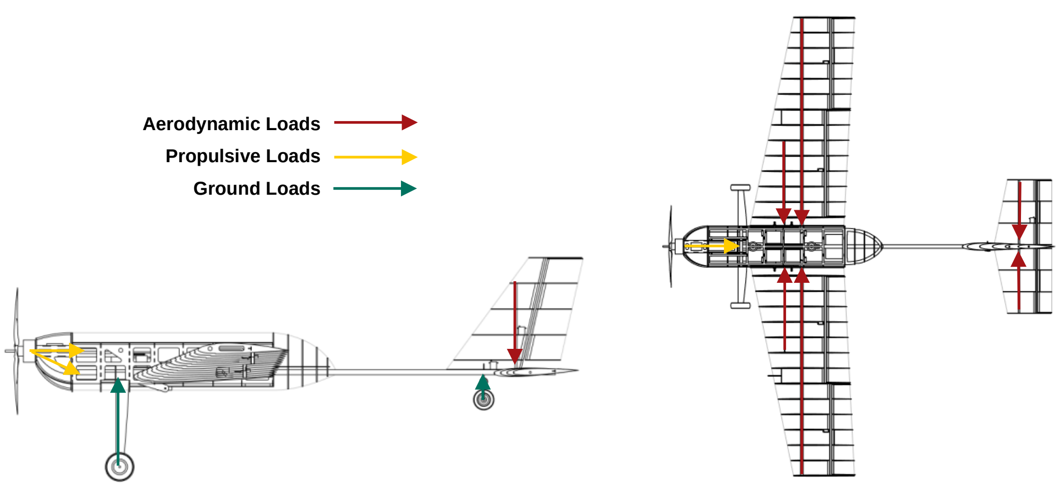


Figure 5.2: Load Path Diagram

5.3 Subsystem Design

5.3.1 Fuselage

The fuselage was designed to house the electronics, carry payloads, and connect to the wing and tail. The main structure is made up of basswood and birch plywood formers as shown in Figure 5.3. A CF rod runs lengthwise through the fuselage to increase stiffness and provide a mounting point for the tail. This is the main CF rod through which all loads are directed to. Plywood panels form the sides of the fuselage, providing an attachment point for the wing and helping to transfer wing loads to the main CF rod. An aluminum rod bent at 10° is also integrated into the fuselage to connect the two wing halves. The top of the fuselage was designed to be detachable for easy access to the battery, payload, and electronics compartments. The bottom of the fuselage acts as a ramp that can be lowered for package deployment in M3. A basswood motor mount runs through the nose and connects to the main lengthwise CF rod as well as an additional horizontal CF rod for added support.

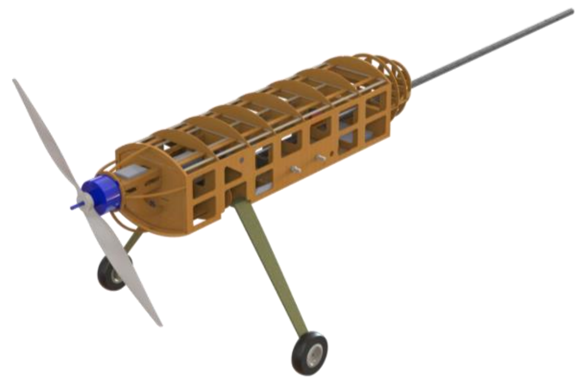


Figure 5.3: Aircraft Fuselage

5.3.2 Wing

Each half-wing is made of 15 ribs with a rear CF spar and a front wooden spar for support. The CF spar was designed to run the full length of the wing to take most of the aerodynamic bending load. The second wooden spar was included to resist twisting of the wing. The spars are angled at 5° to achieve the desired dihedral, and all outer ribs are oriented perpendicular to the spars. The innermost rib on each wing is oriented perpendicular to the ground plane in order to create flush contact with the side panels of the fuselage. The half-wings are designed to be bolted to the side panels. This allows for loads to be transferred into the side panels, while still allowing the half-wings to be detached for transport. The CF spar was designed to extend past the innermost rib to interface with the bent aluminum rod integrated into the fuselage, which provides added support against bending. The leading edge of the wing will be wrapped with balsa wood to increase stiffness and achieve the correct curvature. Servo and control surface attachment points were also included in the wing structure as shown in Figure 5.4 below.

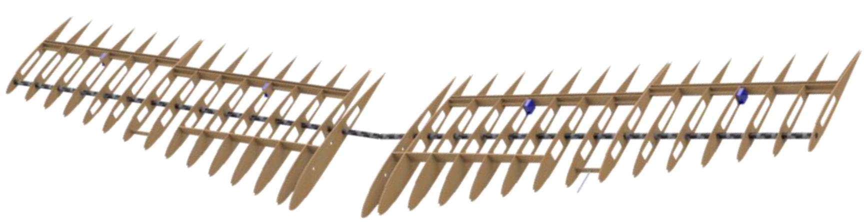


Figure 5.4: Aircraft Wing

5.3.3 Empennage

The empennage was designed using the same techniques as the main wing. The HT and VT are made of ribs supported by a CF spar and a wooden spar. The CF rods intersect directly with the tail boom. This allows tail loads to be transferred directly to the central CF rod. The tailwheel spring also intersects the main CF rod and is secured in place using a collar. Mounting plates for control servos and the rudder and elevator hinges are also integrated into the rib structure. The full empennage structure is shown in Figure 5.5.

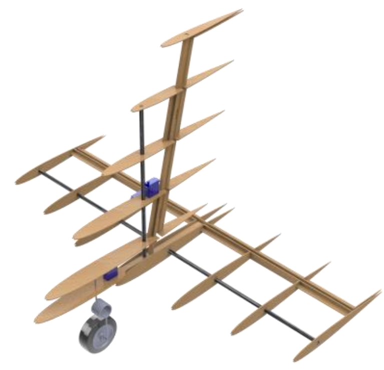


Figure 5.5: Aircraft Empennage

5.3.4 Main Landing Gear

Since WUDBF did not have the equipment to manufacture the main landing gear, a set of CF landing gear was purchased online. The type and placement of the main landing gear were chosen based on three conditions: a takeoff AoA between 12°-15° (to aid in takeoff), a vertical angle with the CG of 16°-25° (based on rules-of-thumb [5]), and a ground clearance of at least 12.5 in (to accommodate the 24 in prop). Based on these criteria, the JTC-S24 standard style CF landing gear was chosen. The main landing gear is supported by two basswood plates on top and bottom. These plates intersect the first three formers to transfer loads to the main fuselage structure. Bolts running through the plates and the landing gear secure the gear to the fuselage. Additional structures that run perpendicular to the basswood plates give added compressive strength as shown in the figure below.

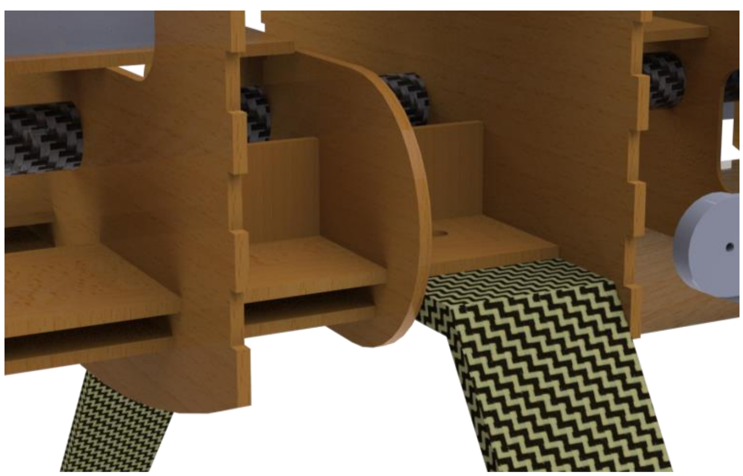


Figure 5.6: Closeup of Landing Gear Attachment

5.3.5 Payload Storage

The payload storage compartment is located within the fuselage just above the wings. This placement was chosen to minimize CG shifts during package deployment in M3. Figure 5.7 shows a close up of the payload compartment filled with four packages. Formers, the fuselage sidewalls, and the top lid contain the packages to minimize shift during flight. Based on empirical testing, WUDBF determined that 10 syringes can fit within the same space occupied by one package. Thus, for M2, the 40 syringes will be split into groups of ten and inserted into the package slots.

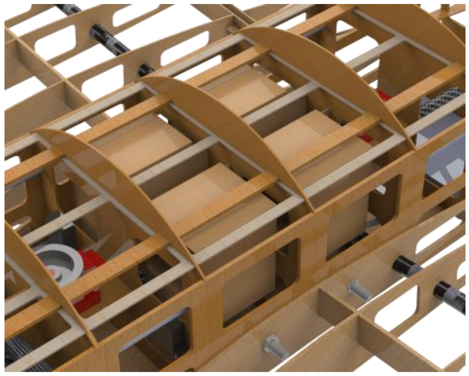


Figure 5.7: Payload Storage Compartment

5.3.6 Package Deployment Mechanism

The package deployment mechanism was designed to be as simple as possible in order to improve reliability and save weight. Because of the need to not trip the 25G shock sensors, the deployment mechanism was designed as a two-stage system with a servo-operated release mechanism and a ramp based lowering mechanism.

The release mechanism consists of servos attached directly to the walls of the payload storage compartments. Aluminum servo arms extend out beneath the packages to support them during flight. Because the packages are supported perpendicular to the servo's axis of rotation, the weight of the package acts primarily on the servo body and mount and not on the servo gears or motor. To release a package, the servo arm is swung 90°, allowing the package to drop down internally onto the ramp. The release stage was designed to be performed with the ramp in the fully retracted position, minimizing the distance the package has to drop.

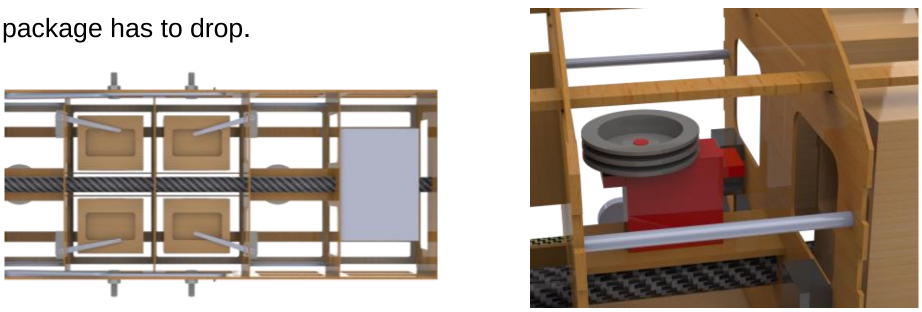


Figure 5.8: Closeups of Release Mechanism from Bottom with Ramp Removed (left) and Lowering Mechanism Pulley Servo (right)

The lowering mechanism consists of a ramp operated by two continuous rotation pulley servos, each connected to two winch lines via dual spools (see Figure 5.8 above). One servo controls the two lines attached to the front of the ramp, and the other controls the two lines attached to the back. This allows the ramp to be lowered straight down parallel to the ground and then tilted slightly at the end so the package can slide gently off the ramp (see Figure 5.8).

Due to concerns about the amount of wind in Wichita, two supporting guide arms were designed to prevent the ramp from twisting or flipping over and acting as a sail. Each guide arm consists of two sections. One section is attached to the side panel of the fuselage and the other to the ramp. During flight, the arms fold up into the fuselage. The two sections were sized such that at maximum extension, the angle between them does not exceed 180°, allowing the arm to fold back up properly into place.

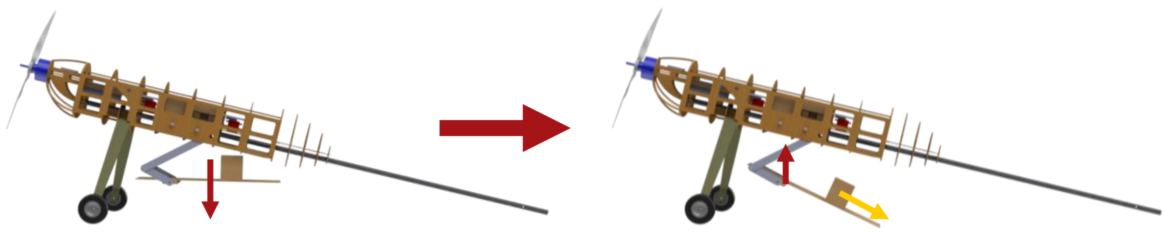


Figure 5.9: Lowering of the Ramp

5.3.7 Propulsion System

The propulsion system consists of a SunnySky X7025-30cc-270 motor spinning an APC 24x12 prop. Two different batteries are used for M2 and M3 in order to optimize performance. For a complete description of the design of the propulsion system, see section 4.5. Static testing in conjunction with flight testing confirmed that the propulsion system met thrust, power, and endurance requirements. The table below shows breakdowns of the propulsion system for each mission.

Table 5.2: Propulsion System Component Breakdown

Component	Mission 1 and 2	Mission 3
Motor	SunnySky X7025-30cc-270 (270 Kv)	
Propeller	APC 24x12	
Propulsion Battery	Turnigy Graphene Panther 3000 mAh 6S 75C	Turnigy Nano-Tech 4500 mAh 6s 45/90C
ESC	AeroStar Advance 150 A HV (6~12S)	
Receiver Battery	Tattu 3S 850 mAh 75C	
Receiver	FrSky X8R 8/16Ch	

5.4 Weight and Balance

The aircraft CAD model in Solidworks was used to determine the CG for each component. The weights were estimated based on component volumes with known densities, measured component weights, and previous WUDBF aircraft. The leading edge of the center of the wing was taken to be the origin, and a mutually orthogonal coordinate system was established with the positive X-axis pointing out the nose of the aircraft, the positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right wing when viewing the aircraft from behind. The table below shows the estimates for each mission.

Table 5.3: Weight and Balance for Each Mission

Component	Weight [kgf]	CG_x [mm]	CG_y [mm]	CG_z [mm]
Wing	1.82	-299.7	0.00	-56.39
Fuselage	1.37	-321.3	0.00	-42.16
VT	0.03	-1390	0.00	-190.5
HT	0.04	-1458	0.00	-26.42
Main Gear	0.34	19.05	0.00	104.6
Tailwheel	0.01	-1267	-12.45	12.45
7x Control Servos	0.09	-833.4	19.56	-72.90
4x Release Servos	0.05	-201.2	0.00	-46.23
2x Pulley Servos	0.11	-204.7	0.00	-64.77
ESC	0.12	257.3	0.00	-123.4
Control Electronics	0.54	-274.1	0.00	-55.63
Propulsion Battery	0.75	125.0	0.00	-74.93
Motor	0.61	350.0	0.00	-100.08
M1				
Overall	**5.68**	**-157.5**	**0.00**	**-35.81**
M2				
Syringes	0.76	-197.1	0.00	-59.94
Overall	**6.44**	**-162.3**	**0.00**	**-38.35**
M3				
Package 1	0.23	-249.9	-80.01	-50.04
Package 2	0.23	-249.9	-80.01	50.04
Package 3	0.23	-150.1	-80.01	-50.04
Package 4	0.23	-150.1	-80.01	50.04
Overall (4 Packages)	**6.59**	**-163.1**	**0.00**	**-41.91**
Overall (3 Packages)	**6.36**	**-160.0**	**2.03**	**-40.39**
Overall (2 Packages)	**6.13**	**-160.5**	**0.00**	**-38.86**
Overall (1 Package)	**5.91**	**-157.2**	**-2.03**	**-37.34**

The vehicle is designed such that the position of the CG can be adjusted using battery placement and ballast, allowing the aircraft to be stable in all flight configurations. For M3, the packages were located in a way that longitudinal CG shifts are minimized following package deployment. The lateral and longitudinal CG shifts are small enough that they can be counteracted with pre-programmed trim that will be determined during test fights.

5.5 Structural Performance

In order to verify structural performance of the aircraft, WUDBF utilized Solidworks finite element analysis (FEA) to simulate the major structural components of the aircraft under their maximum expected load conditions.

5.5.1 Fuselage FEA

To verify the strength of the fuselage design and confirm its ability to withstand aerodynamic loads from the wing, FEA was performed on the main fuselage structure. A load factor of 2.5 on the wing was simulated at the wing's attachment points on each side panel, the two bolt holes and the hole for the CF spar. The resulting stress plot is shown below.

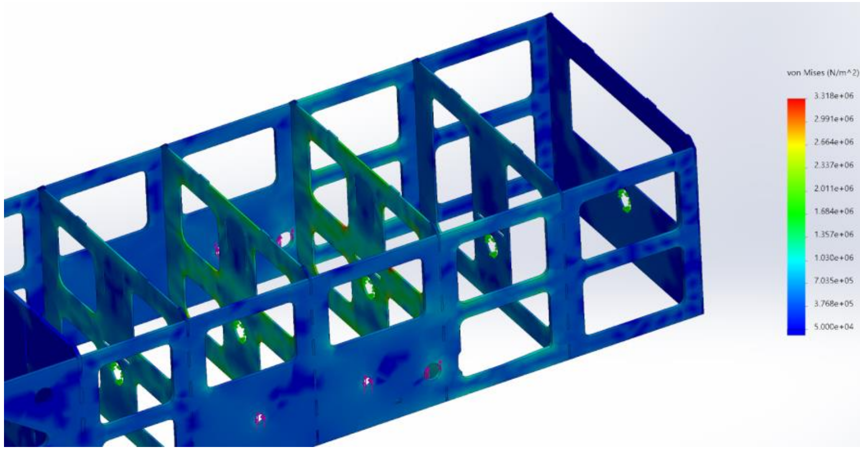


Figure 5.10: von Mises Stress Plot of the Fuselage Experiencing a Load Factor of 2.5

The largest loads ranged from 1-3.3 MPa distributed through the two ribs directly adjacent to the holes. Stress concentrations occurred at the corners of the weight saving cuts, shown in red. Given that the minimum tensile strength of birch plywood, the material selected for the formers, is 27.6 MPa [12], the fuselage structure had an adequate factor of safety of 8.36.

5.5.2 Wing Spar FEA

FEA was performed in Solidworks on the CF spar to ensure the wing could withstand a wingtip loading test (approximate load factor of 2.5). The finite lift distribution was approximated by XFLR5 to be near elliptical. In order to input the distribution into Solidworks, curve fitting was performed in MATLAB to generate a continuous distribution as shown in the figure below. Note that the lift distribution is plotted as a fraction of total lift as a function of unit span.

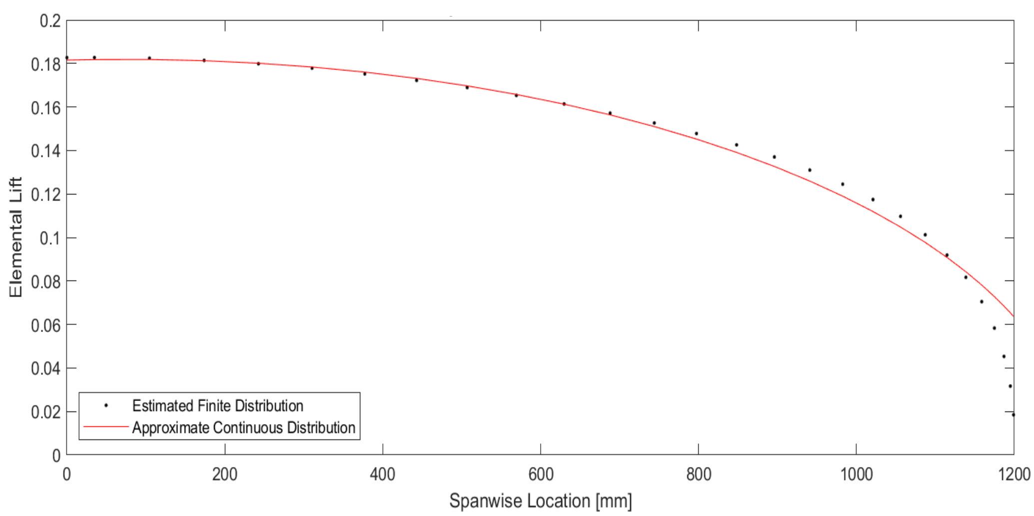


Figure 5.11: Finite and Continuous Spanwise Lift Distribution

Using the continuous lift distribution and applying a load factor of 2.5, the following von Mises stress plot was generated using Solidworks.

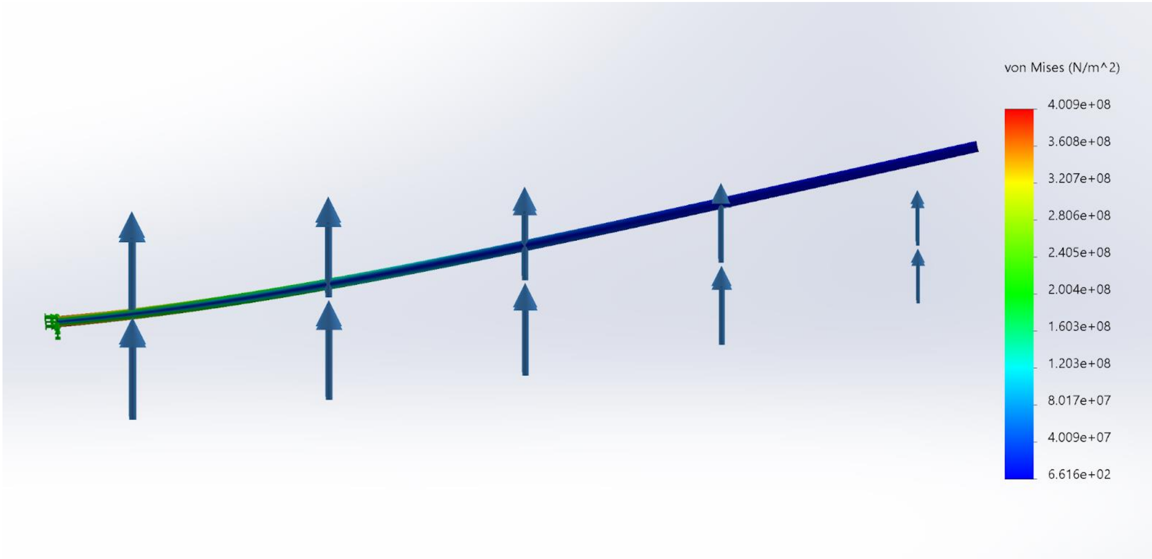


Figure 5.12: Von Mises Stress Plot of Wing CF Spar Experiencing a Load Factor of 2.5

Based on the simulation, the maximum flexural stress experienced by the CF spar is 401 MPa. Given a maximum tensile strength of 1896 MPa from the manufacturer, this results in an adequate factor of safety of 4.7 [13]. Based on the FEA, the 2.5g load case also produced a wingtip displacement of 6.29 in or 8.4°, which was considered reasonable given that the actual displacement of the spar would be lower with the addition of the wooden structure around it.

5.5.3 Main Landing Gear FEA

A FEA was performed on the front landing gear to ensure it had the structural integrity to survive landing. A landing load factor of 3 was applied to the top surface of the gear to model a normal landing, and a von Mises stress plot was generated as shown in the figure below.

As shown in Figure 5.13 the max von Mises stress was 22.1 MPa. Given a maximum tensile strength of 1896 MPa for carbon fiber, this gives a more than adequate factor of safety of 85.8 [13]. Additionally, the maximum displacement experienced by the landing gear was 0.005 in, which was acceptable since there was 6.5 in of designed prop clearance.

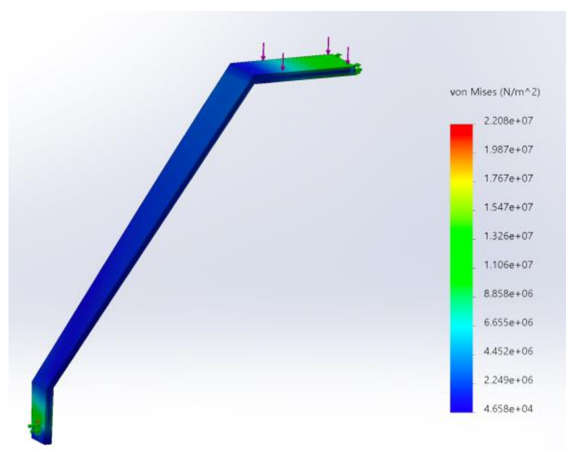


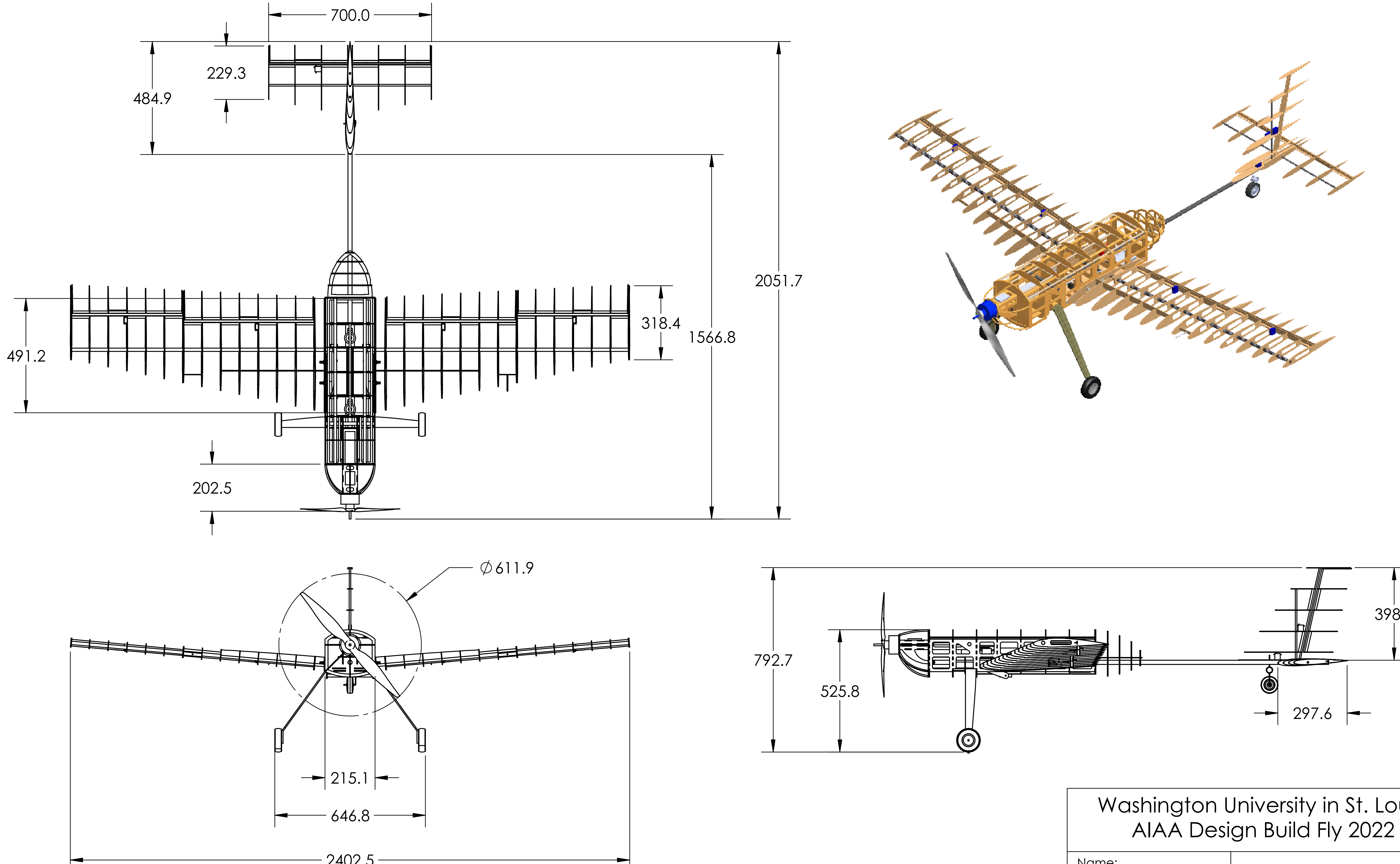
Figure 5.13: von Mises Stress Plot of the Main Landing Gear Experiencing a Load Factor of 3

5.6 Predicted Final Aircraft Performance

Using the updated weights in Table 5.4, Km_sim was used to simulate the expected flight and mission performance of the final aircraft. The extracted performance estimates are shown in the table below. To obtain normalized scores, best scores for M2 and M3 were estimated based on Section 3.3.

Table 5.4: Predicted Final Aircraft Performance

	Mission 1	Mission 2	Mission 3			
Inputs						
All-Up Weight [kgf]	5.68	6.44	6.59	6.36	6.13	5.91
Payload Weight [kgf]	-	0.76	0.92	0.69	0.46	0.23
Wing Loading [kgf/m^2]	5.70	6.47	6.62	6.39	6.15	5.93
$C_{L,max}$	1.44	1.44	1.44			
Static Thrust [kgf]	10.54	10.54	10.54			
Cruise Throttle	1	1	0.7			
Turn Load Factor	2.5	2.5	2.5			
Outputs						
Ground Roll [m]	3.332 (10.93 ft)	3.386 (11.11 ft)	3.427 (11.24 ft)	3.420 (11.22 ft)	3.412 (11.19 ft)	3.405 (11.17 ft)
$C_{L,cruise}$	0.411	0.411	0.411	0.411	0.411	0.411
$C_{D,cruise}$	0.411	0.411	0.411	0.411	0.411	0.411
Cruise Velocity [m/s]	21.83	21.89	20.73	20.69	20.65	20.62
Number of Laps	3	3	4			
Lap Time [s]	47.41	43.84	56.86	56.81	56.73	56.68
Mission Time [min]	2.371	2.192	3.784 (excluding ground phase)			
Scoring						
WUDBF Score	1	18.25	4			
Normalized Score	1	1.616	2.667			



ALL DIMENSIONS ARE IN MILLIMETERS

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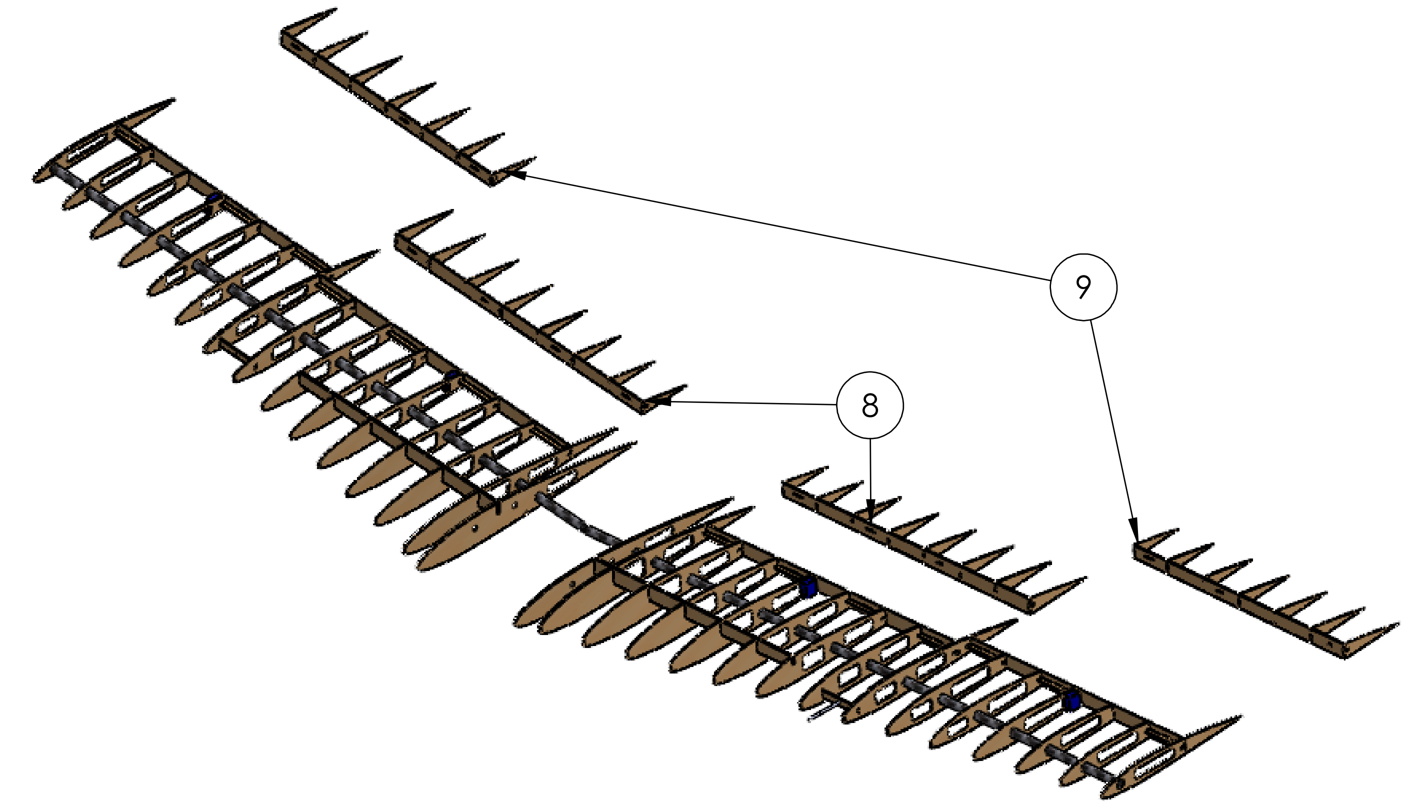
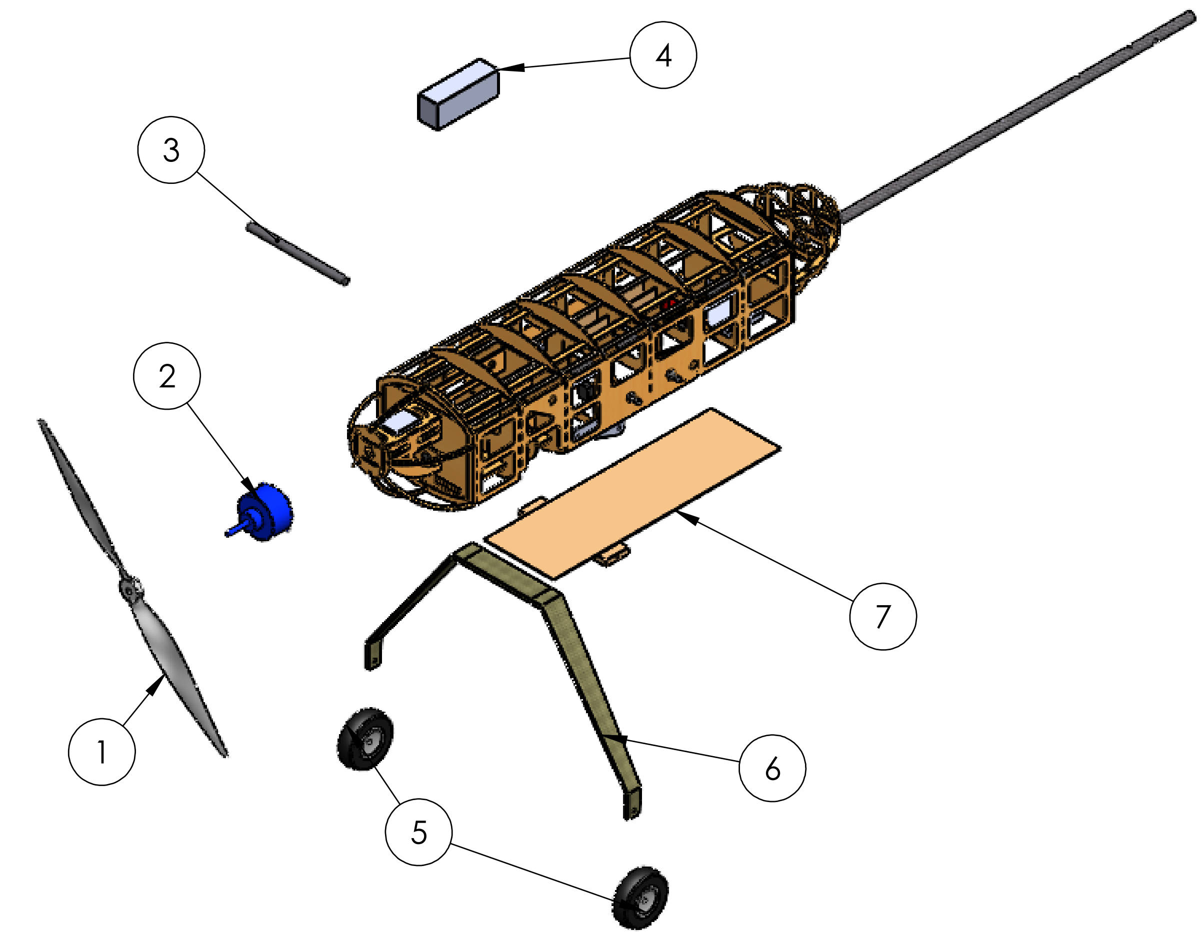
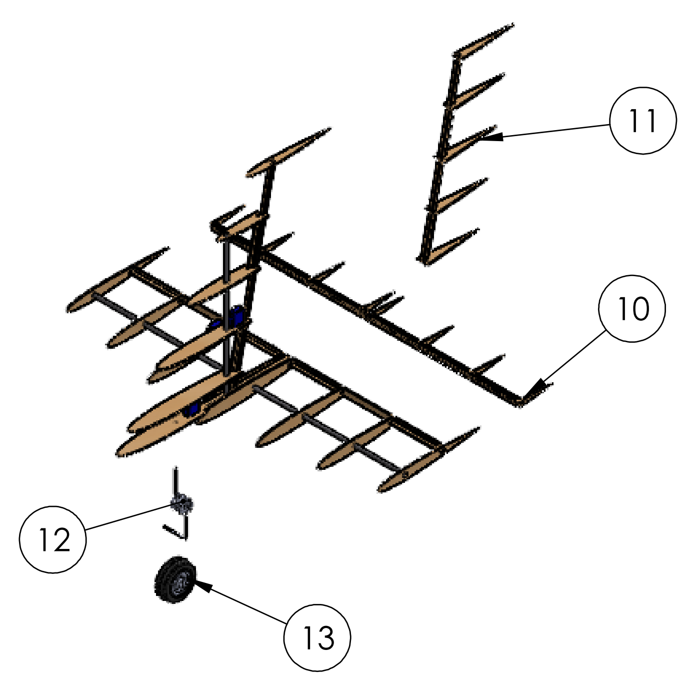
Name: Pflyzer	Aircraft 3-View
SCALE: 1:16	SHEET 1 OF 4







Bill of Materials			
Item No.	Item Name	Qty	Material
1	Propeller	1	Nylon
2	Motor	1	Aluminum
3	Motor Mount Spar	1	CF
4	Battery	1	Lithium Polymer
5	Front Landing Gear Wheels	2	Nylon
6	Front Landing Gear	1	CF
7	Ramp	1	Basswood
8	Flaps	2	Balsa
9	Ailerons	2	Balsa
10	Elevator	1	Balsa
11	Rudder	1	Balsa
12	Rear Landing Gear Spring	1	Aluminum
13	Rear Landing Gear Wheel	1	Foam

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Name: Pflyzer

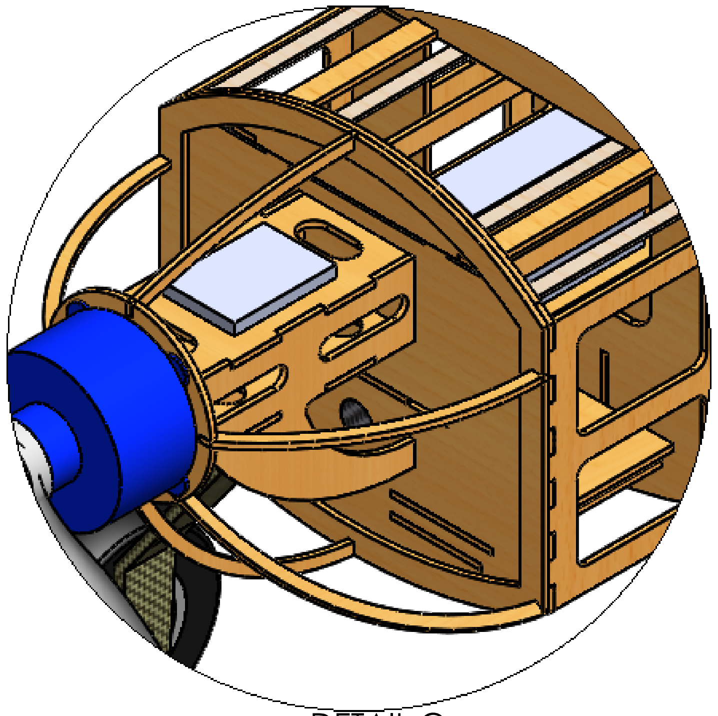
Aircraft Exploded Layout

SCALE: 1:12

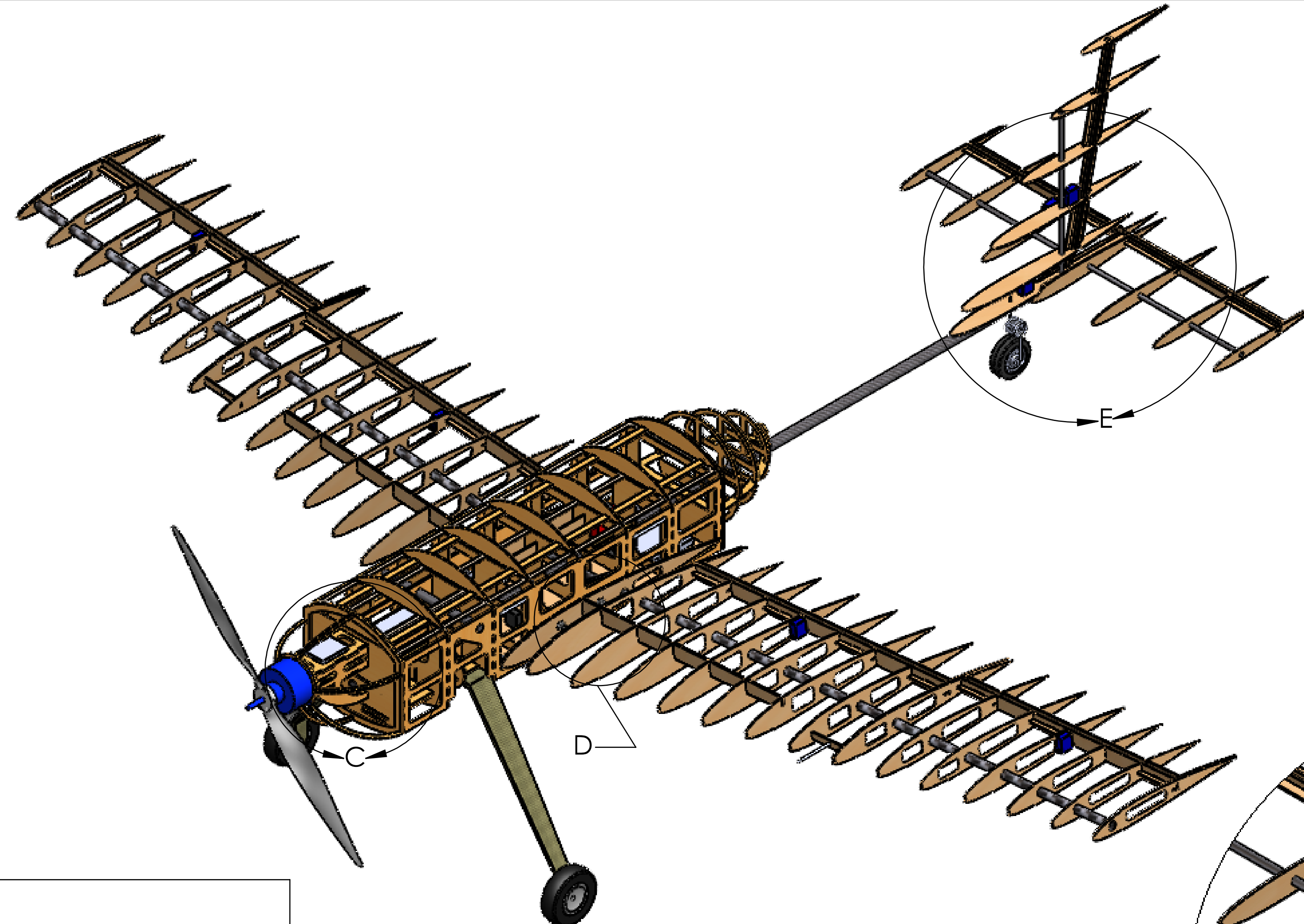
4 3 2 1

B

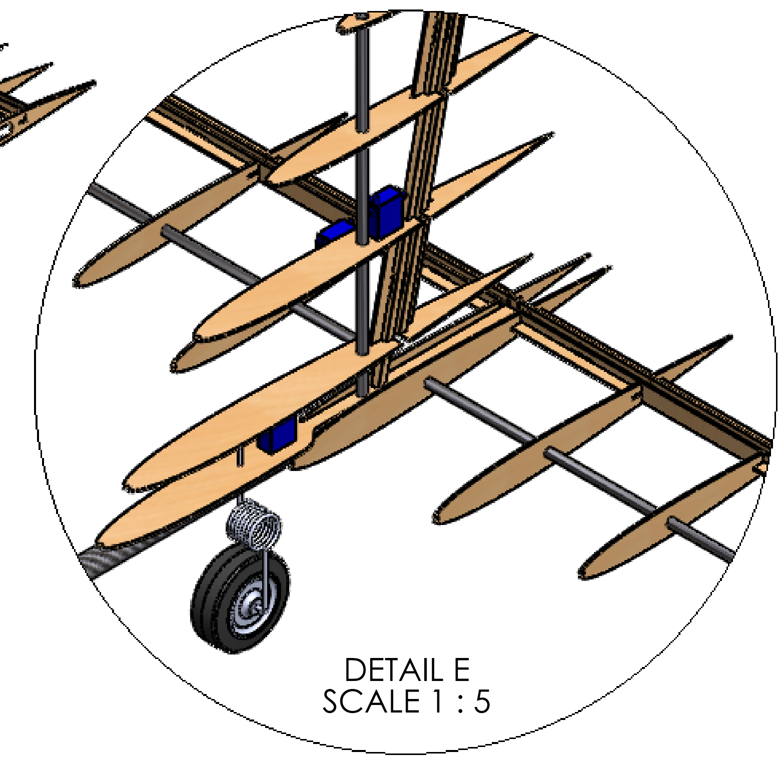
MOTOR MOUNT STRUCTURE



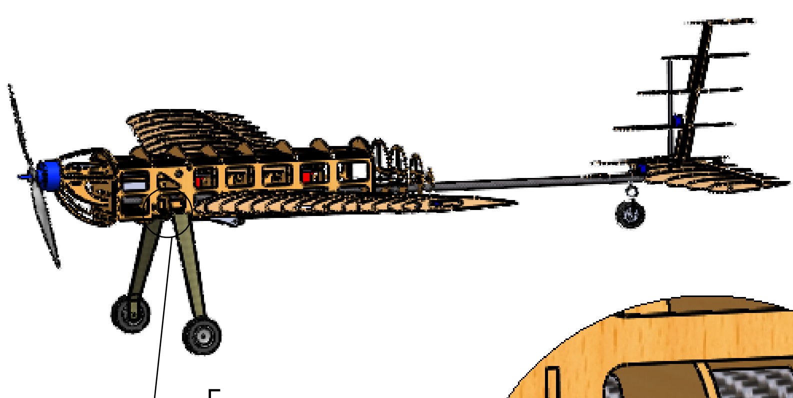
DETAIL C
SCALE 1 : 3



EMPENNAGE CONNECTION

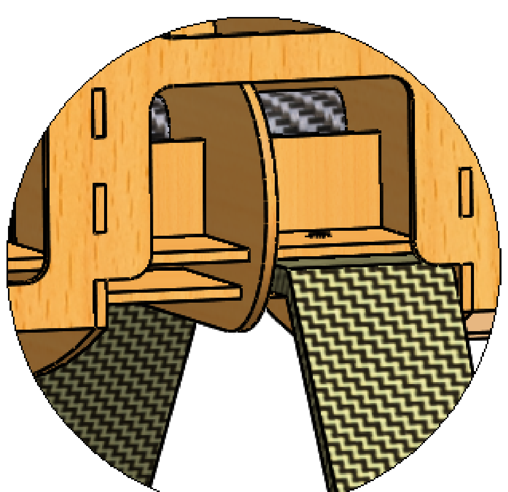


DETAIL E
SCALE 1 : 5



SCALE 1 : 20

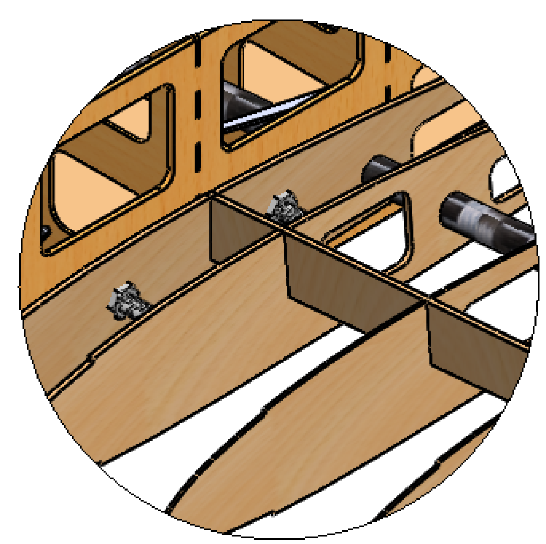
A



LANDING GEAR STRUCTURE

DETAIL F
SCALE 1 : 2

WING-FUSELAGE CONNECTION



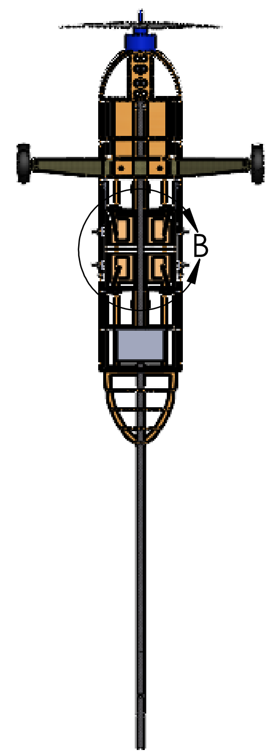
DETAIL D
SCALE 1 : 3

Washington University in St. Louis AIAA Design Build Fly 2022	
Name: Pflyzer	Aircraft Detail
SCALE: 1:10	SHEET 3 OF 4

4 3 2 1

B

PACKAGE RELEASE MECHANISM
(BOTTOM VIEW, RAMP HIDDEN)



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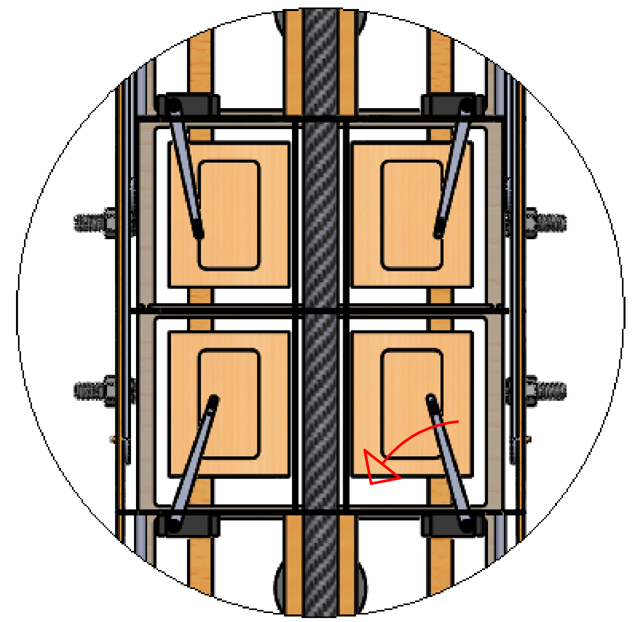
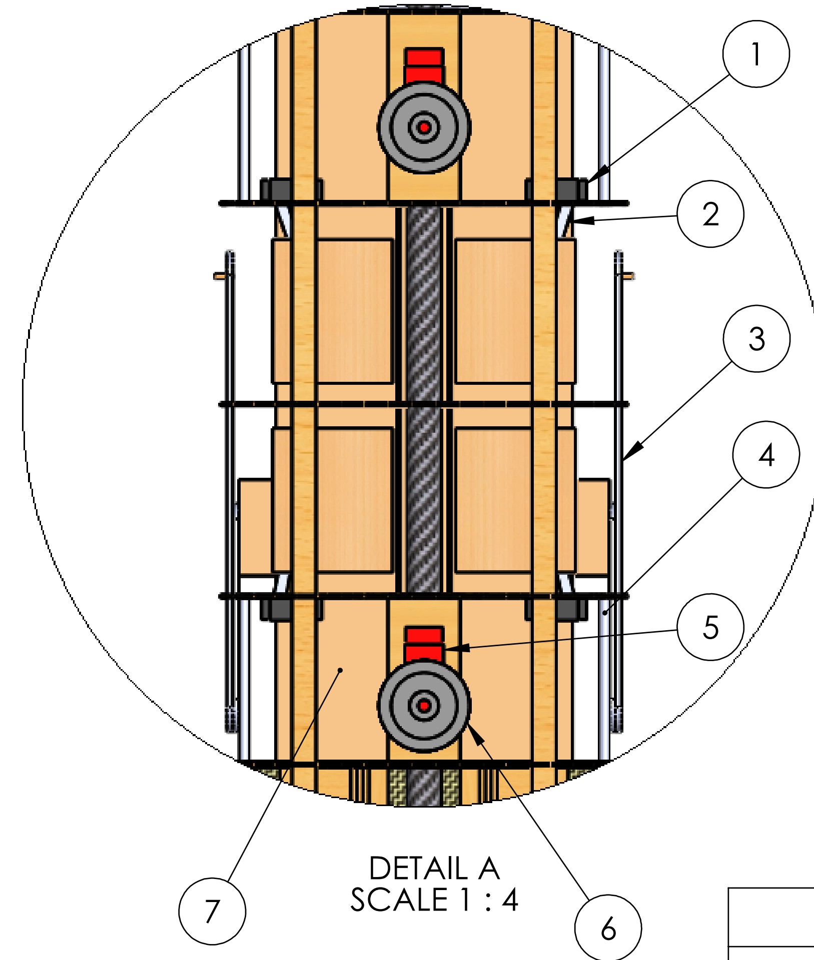
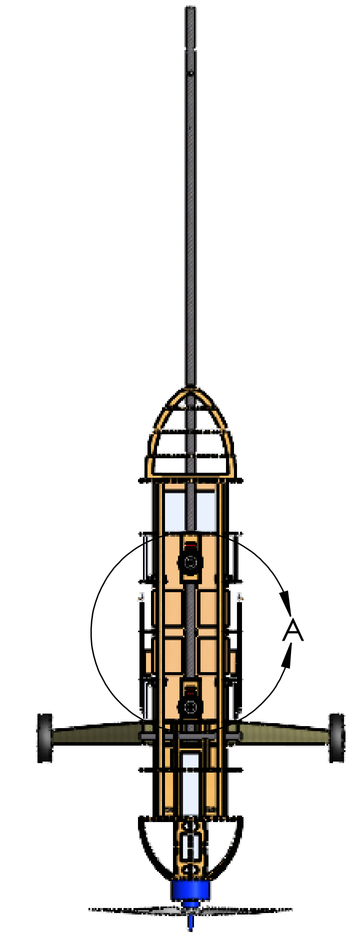


DETAIL B
SCALE 1 : 4

DEPLOYMENT MECHANISM LIST OF MATERIALS
(TOP VIEW, SIDE WALLS HIDDEN)

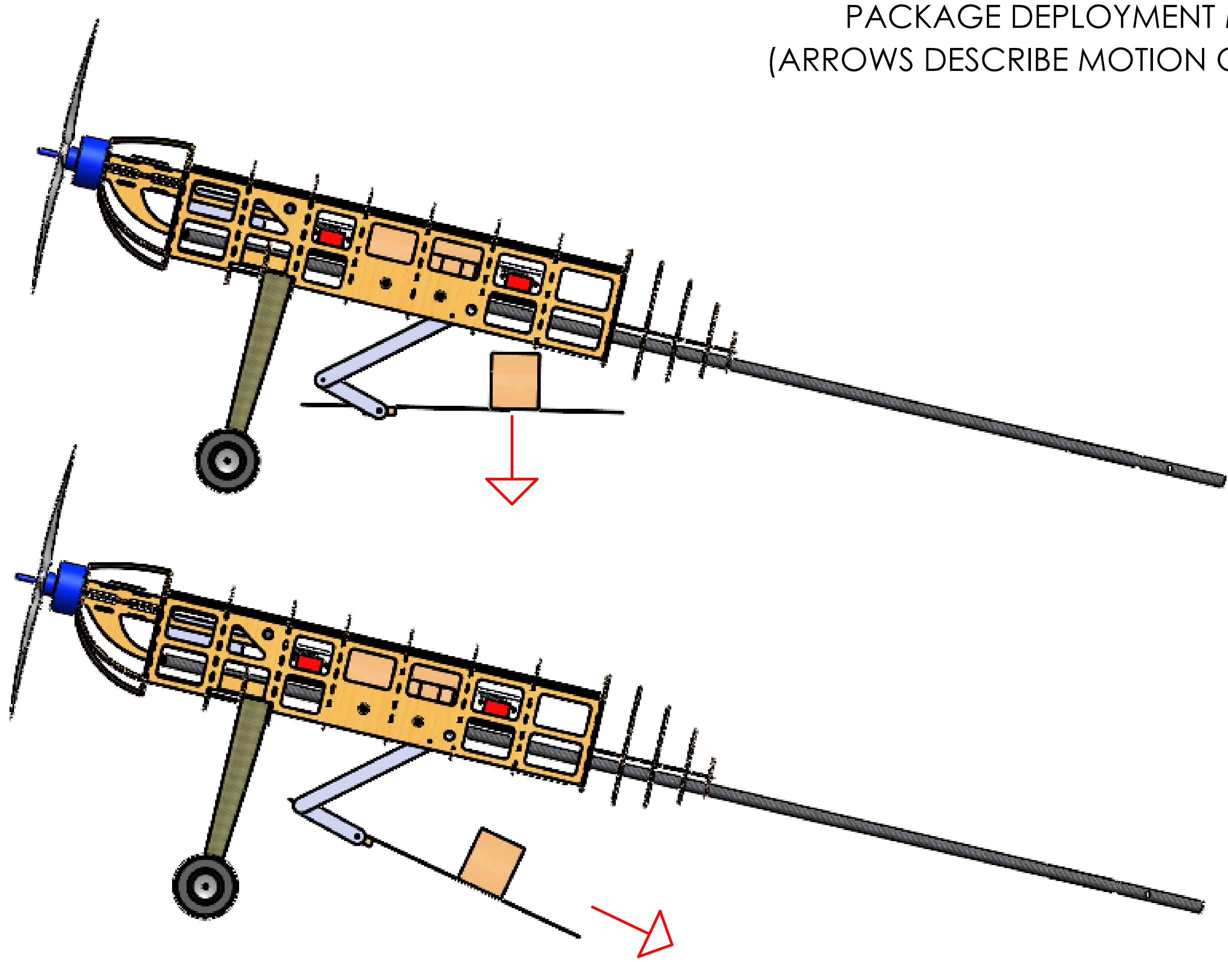


DETAIL A
SCALE 1 : 4



SCALE 1 : 16

PACKAGE DEPLOYMENT MECHANISM
(ARROWS DESCRIBE MOTION OF THE PACKAGE)



A

Bill of Materials		
Item No.	Item Name	Qty
1	Package Release Servo	4
2	Package Release Servo Arm	4
3	Deployment Guide Arm	2
4	V Groove Pulley Rod	4
5	Winch Pulley Servo	2
6	Winch Pulley	2
7	Ramp	1

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AIAA Design Build Fly 2022

Name: Pflyzer	Payload Accomodation
SCALE: 1:12	SHEET 4 OF 4

6. Manufacturing Plan

6.1 Selection of Manufacturing Processes

Multiple manufacturing processes were considered to identify the best method for constructing the aircraft and the various subsystems.

6.1.1 Selection Criteria

Key evaluation criteria included ease of construction, effectiveness, feasibility, and weight. Materials which had been used in past aircraft were generally preferred, given manufacturing methods for those materials had already been refined. Finally, cost was also considered due to WUDBF's limited budget.

6.1.2 Solid Foam

Solid foam components could be designed and fabricated quickly since there are no internal bracing structures. However, achieving accurate shapes in foam requires using a hot-wire cutter, something WUDBF has little past experience with. It would be necessary to design and build a hot wire cutter and the necessary jigs, which would increase the time needed to build the aircraft. Although foam is light and has sufficient strength when paired with CF or fiberglass, the time constraints meant it was not chosen.

6.1.3 Additive Manufacturing

3D printing has advantages during construction due to the ease of fabricating structures as single components and the streamlined workflow from CAD to printer. However, 3D printers are slower than other methods for constructing large shapes. 3D printed parts are also heavier and weaker than those made from other materials, especially in shear along the seams between layers. While WUDBF ultimately did not make use of 3D printed parts for the main aircraft structure, more intricate components that don't bear significant loads such as servo mounts were 3D printed due to the flexibility and ease of 3D printing.

6.1.4 Composite Layup

Fiberglass or CF composite layup methods were considered due to their high strength and low weight. These methods however require construction of molds or plugs, which were determined to be too time-consuming and difficult to manufacture with current resources. WUDBF also has limited experience with these methods. That being said, prefabricated CF rods and fiberglass layups were used to reinforce the major load bearing components on the aircraft.

6.1.5 Wood Construction

Wood construction is the manufacturing technique WUDBF is most experienced with. WUDBF has a laser cutter that can be used to rapidly cut out wooden parts. Balsa, basswood, and birch ply are the primary woods used for RC aircraft, and each have different material properties. This offers flexibility when choosing where weight can be saved or where extra strength is needed. Assembly of wooden components can be done using CA glue and epoxy. CA glue dries quickly and forms a strong bond, making it the primary adhesive in wooden construction. Two-part epoxy offers a stronger alternative for components that bear more structural loads. In order to maintain the aerodynamic shape of the aircraft, the frame can be covered in Monokote, a heat shrinking plastic film that can be tacked onto the frame of the aircraft then heated to pull it tight across the aircraft. Monokote is relatively thin and susceptible to tearing but can be quickly and easily repaired.

6.1.6 Processes Selected for Major Components

Wood construction, with Monokote coating, was chosen as the primary manufacturing method for all major components due to its structural qualities and WUDBF’s previous experience. All wooden components including wing ribs and fuselage formers are cut using a laser cutter. These pieces are then assembled using CA and epoxy. CF parts and fiberglass reinforcement are added in key areas for structural support. In particular, the central rod, wing spars, and landing gear are CF parts purchased online, and the wood plates connecting the landing gear to the fuselage are reinforced with fiberglass.

6.2 Detailed Manufacturing Processes

The manufacturing processes used in the construction of the aircraft are detailed below. Note that these descriptions are primarily based on the construction of the prototype aircraft and some adjustments may occur for the final aircraft, which has yet to be built.

6.2.1 Fuselage

The fuselage is constructed primarily from birch ply formers, two side panels also made from birch ply, and a mix of basswood and balsa stringers. It was found that balsa stringers warp under the force of the Monokote, so basswood stringers are used at sharp corners. During construction a jig is used to align the formers. This jig consists of a wooden plate with laser cut holes. Jig posts attached to the formers slot into the holes. With the formers properly aligned in the jig, the side panels are attached and stringers CA glued into place. The central CF rod is then inserted and epoxied to the formers. Two basswood landing gear plates are then installed. These plates are reinforced with fiberglass for additional strength. The CF landing gear is sandwiched by these plates and secured using bolts. The nose containing the motor mount is constructed separately from the main fuselage. The nose slides into the main fuselage body and is secured via a horizontal CF rod epoxied to the side panels and a bolt running through a hole drilled in the central CF rod. Following construction of the frame, the entire structure is then covered in Monokote.

6.2.2 Wing

The wing is composed of ribs, stringers, a main CF spar, and a secondary wooden spar. Each half-wing is constructed separately and is designed to be easily removable. The ribs are aligned using slotted basswood jigs, with each rib fitting into a designated slot. Once the ribs are aligned, the CF and wooden spars were inserted and glued using epoxy. A quarter inch square basswood stringer is then attached to the leading edge of the wing and sanded down to achieve the correct curve. Next, the mounting plates for the control surfaces are installed, the jigs cut off and sanded down, and the wing flipped over. Basswood stringers are then added. The leading edge is then formed by wrapping balsa sheets soaked in warm water around the leading edge stringer. 3D printed servo mounts are then glued to the ribs.

Control surfaces are constructed in the same manner as the wing. Hinges are glued to the control surfaces and then to the mounting plates in the wing to secure them in place. Finally, the wing and control surfaces are covered with Monokote.

6.2.3 Wing Attachment

Each half-wing is attached to the fuselage via the side panels. The innermost rib of each half-wing is offset at 5° from the vertical to give the desired wing dihedral when mounted flush with the side panels. Holes in the innermost ribs line up with holes in the side panels, through which bolts are used to secure the wings in place. In addition, an aluminum tube bent to 10° manually while checking with an angle gauge is integrated into the fuselage. The CF spars in the wings slide around this tube to give added support help align the wings.

6.2.4 Empennage

The prototype tail is constructed using basswood stringers, balsa wood, and thin CF rods. Like in the wing, basswood stringers serve as the leading edges. These stringers are sanded to the shape of the desired airfoils. Each component of the tail is constructed separately in an identical manner to the wings, using similarly designed jigs. The elevator and rudder are also constructed separately and then attached to the HT and VT respectively using hinges and CA glue. To connect the tail to the fuselage, holes oriented perpendicularly to each other are drilled in the central CF rod, which extends out of the fuselage and acts as the tail boom. A custom hole drilling jig is used to ensure that the holes are perpendicular to each other and aligned properly with the wings and the rest of the aircraft. Thinner CF rods are then inserted into the holes. The VT and HT are then slid over these CF rods and epoxied into place. Following installation of servos and servo mounts, the tail is then covered in Monokote.

6.2.5 Deployment Mechanism

Figure 6.1: Pulley Servo with Dual Spool

Owing to the simplicity of the deployment mechanism, no intricate, custom components are required. The guide arms and ramp are laser cut out of wood. The guide arms are connected to each other, the fuselage, and the ramp using bolts. The release servos are fixed to the sides of the payload compartments using the same 3D printed mounts described previously. Aluminum servo arms purchased online are attached to the release servos to support the package during flight and release the packages during the deployment phase. High torque, continuous rotation servos are used for the ramp pulleys and are mounted in front and behind the payload compartment on wooden plates epoxied to the central CF rod. The dual spools attached to the pulley servos can be purchased online. Two synthetic cables are then connected to each spool, with each cable then running over a guide rod mounted between the formers and down to the ramp where they are tied and CA glued into place.

6.3 Manufacturing Milestones

A manufacturing schedule was implemented to ensure that components were constructed on time. The construction of the prototype and final aircraft were divided into important subtasks which were then used to create the schedule shown in Figure 6.2 below.

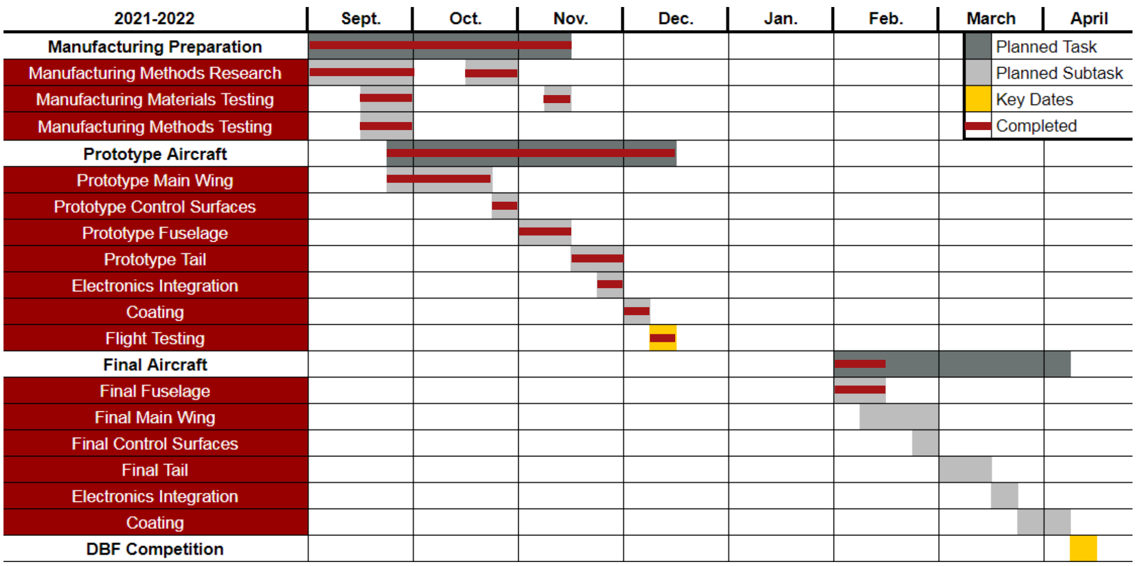

Figure 6.2: Gantt Chart of Manufacturing Schedule for The Prototype and Final Aircraft

7. Testing Plan

Ground and flight testing of the aircraft were performed to validate the design and drive iterations toward the final aircraft design.

7.1 Overall Test Objectives

Primary test objectives are listed in the table below by category. These objectives were achieved through a number of ground and flight tests described in Sections 7.2 and 7.3 respectively.

Table 7.1: Breakdown of Test Objectives

Category	Objective
Propulsion	Collect data from static thrust tests to confirm propulsion performance on the ground. Validate the propulsion system in the air during flight tests to confirm 25 ft takeoff capability, cruise velocity, and endurance.
Manufacturing	Investigate manufacturing processes, in particular the benefits of fiberglassing wood components.
Structures	Conduct wingtip loading tests at maximum AUW to verify structural integrity. Perform torsion and flutter tests to ensure that deflections are not excessive. Confirm integrity of the landing gear via ground tests.
Takeoff	Confirm aircraft can takeoff within 25 ft.
Deployment	Confirm reliability of deployment mechanisms and ensure that shock sensors are not tripped during deployment.
Performance	Record flight data for analysis to confirm performance requirements are met. Obtain feedback from pilot regarding handling qualities.

7.2 Ground Testing

7.2.1 Propulsion Test

Static thrust testing was performed on the chosen propulsion system using an RCbenchmark Series 1585 Dynamometer mounted to a thrust stand. Using RCbenchmark software, thrust, RPM, current draw, and electrical power measurements were collected at incremental throttle levels. The data were then used to validate predicted performance and confirm the propulsion system met the necessary requirements.

7.2.2 Fiberglass Test

WUDBF has had issues in previous years with basswood and/or birch ply landing gear mounting points failing on landing. Thus, it was investigated whether fiberglassing these wooden mounting points could significantly increase strength without excessively increasing weight. Square pieces of balsa, basswood, and birch ply were prepared. Half of the pieces were fiberglassed using 6 oz fiberglass. The test squares were then set up as simply supported beams with force applied to the center of the span, simulating the upwards load resulting from the landing gear hitting the ground. Each square was tested until failure.

7.2.3 Wingtip Test

A wingtip loading test was performed at maximum takeoff weight to validate the structural integrity of the fuselage. This test simulated a 2.5g load case and also verified that the aircraft could survive the wingtip test required at competition. Results from the wingtip test were compared against the FEA performed in Section 5.5.2 to further validate the design.

7.2.4 Structural Motor Thrust and Torque Test

The aircraft structure was tested to verify that it could withstand thrust and torque loads from the motor. Thrust loads were simulated by attaching a spring scale to the motor and pulling at a force of 15 kgf to simulate the expected maximum static thrust of 10.54 kgf times a 1.5 factor of safety. To simulate motor torque, a clamp was fixed to the motor. A spring scale was then attached to the end of the clamp and a 6 N·m torque was applied equivalent to the expected torque of 4 N·m times a 1.5 factor of safety.

7.2.5 Flutter Test

Torsion tests were performed on both the wing and tail to gauge flutter risk, which is an issue WUDBF has had in the past. To conduct the test, a reasonable torque was applied and then quickly released. The resulting vibration was observed for both the magnitude of vibration and the damping characteristics. Since WUBDF did not possess the accelerometers needed to conduct this test quantitatively, qualitative judgments were made to verify that the vibration was broadly acceptable based on prior experience.

7.2.6 Deployment

In order to verify the design of the deployment mechanism, mockups of both the release and lowering mechanisms were built to test their operation. For the release mechanism, the primary objective was to test whether the servo arm could support the weight of the package, and whether the package could be reliably released. For the lowering mechanism, the primary objective was to confirm that the package could be dropped the short distance required to hit the ramp after internal release and then lowered to the ground without tripping the shock sensors. The maximum height the package can be dropped without tripping the sensor was measured (see Figure below). The maximum ramp angle for which the package could be slid down without tripping the sensor was also measured (see Figure below).

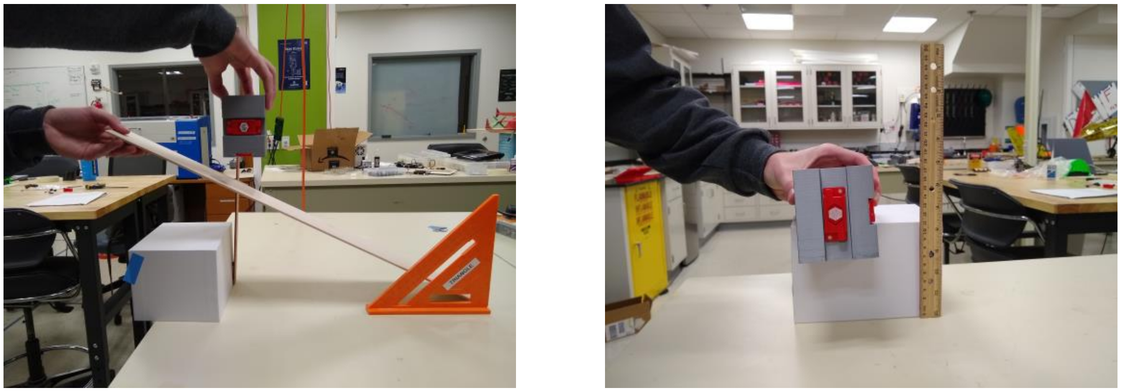

Figure 7.1: Ramp Test Setup (left) and Drop Test Setup (right)

7.3 Flight Testing

Multiple test flights were conducted to verify that the aircraft met design goals. A Pixhawk flight computer was used to capture telemetry during each flight test (see figure below). This data was then analyzed to extract measures of aircraft performance including speed, endurance, and lap times. Feedback from the pilot was also used to evaluate the stability and flight characteristics of the aircraft. Conclusions from flight testing were used to inform changes to the design going from the prototype to the final aircraft. For the full details of each flight test, see the test schedule in Section 7.4.

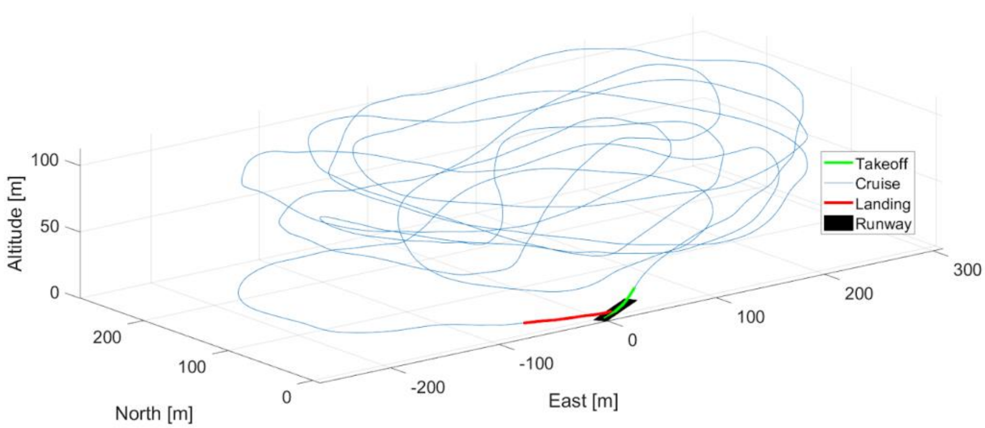

Figure 7.2: Sample Telemetry Data from Test Flight

7.3.1 Ground Inspection and Pre-Flight Checklists

Table 7.2: Ground Inspection Checklist

Component	Tasks
Motor	☐ Motor secured ☐ Propeller free from damage
Fuselage	☐ No cracks or tears ☐ Electronics are connected and secured
Wing	☐ No cracks or tears ☐ Bolted to fuselage ☐ Wingtip test
Payload	☐ Servos connected to receiver ☐ Deployment mechanism moves correctly ☐ Payloads are positioned correctly and secure
Control Surfaces	☐ No cracks or tears ☐ Servos connected to receiver ☐ Control surfaces can deflect fully
Empennage	☐ No cracks or tears ☐ Control surfaces move without interference
Landing Gear	☐ Front wheels secured ☐ Rear wheel alignment check

Table 7.3: Pre- and Post-Flight Checklists

Flight Checklists
Pre-Flight
☐ Payloads secured ☐ Propulsion battery level check ☐ Receiver battery level check ☐ CG position confirmed ☐ Wingtip test ☐ Receiver turned on ☐ Deployment mechanisms armed ☐ Control surfaces test ☐ Landing gear steering test ☐ Range Check ☐ Motor run-up ☐ Fuselage lid is attached and secure
Post-Flight
☐ Throttle down ☐ Arming plug removed ☐ Propulsion battery disconnected ☐ Receiver battery disconnected

7.4 Testing Schedule

Table 7.4: Testing Schedule

Date	Type	Category	Objective
Prototype			
10-16-2021	Ground	Manufacturing	Test fiberglassing of wooden components.
11-20-2021	Ground	Propulsion	Static thrust test to confirm propulsion performance.
11-27-2021	Ground	Structures	Flutter test to check structural stability.
12-12-2021	Ground	Structures	Wingtip test to confirm structure.
12-12-2021	Flight	Performance	Shakedown flight. Test Prototype design. (CRASHED)
Rebuilt Prototype			
1-29-2022	Ground	Deployment	Validation of mock-up deployment design.
2-18-2022	Ground	Structures	Thrust and torque test to ensure verify structure.
2-22-2022	Flight	Performance	Shakedown flight. Test Prototype design.
2-22-2022	Flight	Performance	Full M1 simulated flight. Confirm TOFL.
2-22-2022	Flight	Performance	Full M2 simulated flight. Confirm performance predictions
2-22-2022	Flight	Performance	M3 simulated repeated takeoff-landing with no deployment.
Final Aircraft (Planned)			
3-25-2022	Ground	Performance	Ground mission test.
3-26-2022	Flight	Performance	M1 simulated flight. Evaluate performance.
3-26-2022	Flight	Performance	M2 simulated flight. Evaluate performance.
3-26-2022	Flight	Performance	M3 simulated flight. Test deployment. Evaluate performance.
4-2-2022	Flight	Performance	Practice for competition.
4-9-2022	Flight	Performance	Practice for competition.

8. Performance Results

8.1 Ground Test Results

8.1.1 Propulsion Test Results

Results from the static thrust test confirmed that the propulsion system generally met the desired performance requirements. Test data is compared to eCalc predicted performance in the table below.

Table 8.1: Predicted vs. Actual Performance

Parameter	Prediction	Test Result
RPM	4836	4953
Thrust [Kgf]	10.821	10.05
Max Current [A]	112.94	120.84
Electrical Power [W]	2258.5	2658.48

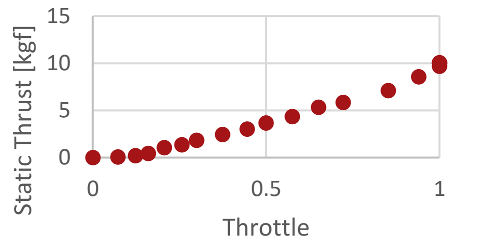

Figure 8.1: Static Thrust Over Increasing Throttle

Based on the static testing, the motor drew slightly more burst current than predicted. This is not an issue, however, since the ESC was sized with significant margin. The efficiency of the actual propulsion system appeared to be slightly lower than predicted, however the static thrust test was conducted indoors, and obstructions and ground effects may have contributed to that decrease in efficiency. Later flight testing confirmed that the propulsion system delivered the desired takeoff capability and cruise performance.

8.1.2 Fiberglass Test Results

Fiberglass testing revealed that fiberglassing basswood increased the strength-to-weight ratio from 101.9 to 225.2 and fiberglassing birch ply increased the strength-to-weight ratio from 164.5 to 311.5. Negligible increases in thickness were observed after fiberglassing. Based on this testing, WUDBF decided to construct the landing gear mounts out of fiberglass reinforced birch ply.

8.1.3 Wingtip Test Results

The plane was supported successfully by just the wingtips. The measured deflection was around 5° in each wing. This agrees with the FEA performed on the CF rod alone in Section 5.5.2 (predicted deflection of 8.4°) since the additional structural elements in the wing provide additional bending support.

8.1.4 Structural Motor Thrust and Torque Test Results

The aircraft structure was able to successfully withstand both the expected thrust and torque loads (times a factor of safety of 1.5).

8.1.5 Flutter Test Results

WUDBF determined that the tail structure was sufficiently strong and would not experience excess fluttering. The torsional rigidity of the fuselage-wing structure, however, was determined to be insufficient and the flutter risk was judged to be beyond acceptable limits. Changes for the final aircraft including changing former material from basswood to birch ply and increasing the size of the side panels were made to improve rigidity.

8.1.6 Deployment Test Results

Testing of the release mechanism determined that an aluminum servo arm could easily support the weight of the package. Since the weight of the package is directed downward, the motor and gears in the servo take minimal load and the majority of the load is taken by the servo body and motor mount. The package drop test determined that the package could be dropped a maximum distance of 3 cm before the sensors are tripped. This is greater than the designed internal distance between the package compartment and the ramp, which validates the design. The ramp test determined that the maximum angle at which the package could be safely slid to the ground without tripping the sensors was 20°. The results of this testing showed that a conventional ramp with one side anchored in the fuselage would not work since the smallest ramp angle that could be achieved was around 30°. This necessitated the twin winch design so that the ramp could be lowered directly downwards allowing for a less steep ramp angle.

8.2 Flight Testing

8.2.1 Dec. 12th, 2021 – Shakedown Flight (Prototype)

The servo wire connecting the elevator servo disconnected just after takeoff. After losing elevator control, the pilot was able to complete one controlled 360° turn before crashing. No flight test data was captured due to the crash. The wing remained largely undamaged, which verified its strength. The fuselage sheared in half longitudinally, with each former snapping in half at the connection (side panels on the prototype extended only halfway up the side of the fuselage). This failure motivated a number of structural redesigns including changing the material of the formers from basswood to birch ply for its stronger, uniform material properties. The side panels for the final aircraft were also increased in height to extend all the way to the top of the fuselage to minimize stress concentrations.

8.2.2 Feb. 22nd, 2022 – Multiple Flights (Rebuilt Prototype)

Table 8.2 summarizes the outcomes of the Feb 22nd test flight.

Table 8.2: Test Flight Outcomes

Flight Description	Result
Shakedown	The aircraft survived the shakedown test. The pilot trimmed the aircraft and was able to achieve stable flight characteristics.
Full Simulated M1	The aircraft was able to takeoff in 10 ft. The mission was completed in 2.6 minutes, which is well within the 5-minute window.
Full Simulated M2	The aircraft was again able to takeoff in about 10 ft. The mission was completed in 2.3 minutes. Max cruise velocity was measured at 21.7 m/s.
M3 Repeated T-L	The aircraft was able to takeoff multiple times within 25 ft while at the fully loaded weight on the same battery. Pilot reported good ground handling characteristics but commented that the landing gear wheels were too hard, leading to bouncing on landing.

During post-flight interviews, the pilot rated the aircraft a 2 on the Cooper-Harper handling qualities rating scale (HQRS). This confirmed that the aircraft's handling characteristics are satisfactory without improvement. Table 8.3 details the actual aircraft performance when compared to predicted performance.

Table 8.3: Predicted Performance versus Actual Performance

Flight Description	Predicted	Actual	Difference
M1 Time [min]	2.371	2.6	-9.2 %
M2 Time [min]	2.192	2.3	-4.8 %
M1 Cruise Velocity [m/s]	21.83	21.7	-0.6 %
M2 Cruise Velocity [m/s]	21.89	21.7	-0.9 %
M3 Cruise Velocity [m/s]	20.67	-	N/A
M3 Battery Endurance [min]	5	6.2	+21.4 %
TOFL (ft)	11	10	-9.5 %

The prototype aircraft generally met or exceeded all performance requirements.

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